

# Optimization for Learning

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# Optimization Problem

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i \in \mathbf{N}_m \\ & h_i(x) = 0, \quad i \in \mathbf{N}_p \end{array} \quad (1)$$

|                                                                                                           |                                        |
|-----------------------------------------------------------------------------------------------------------|----------------------------------------|
| $f_0 : \mathbf{R}^n \rightarrow \mathbf{R}$                                                               | <i>objective function</i>              |
| $f_i : \mathbf{R}^n \rightarrow \mathbf{R}, i \in \mathbf{N}_m$                                           | <i>inequality constraint functions</i> |
| $h_i : \mathbf{R}^n \rightarrow \mathbf{R}, i \in \mathbf{N}_p$                                           | <i>equality constraint functions</i>   |
| $\mathcal{D} = \text{dom } f_0 \cap \bigcap_{i=1}^m \text{dom } f_i \cap \bigcap_{i=1}^p \text{dom } h_i$ | <i>domain</i>                          |

$$f(x) = (f_1(x), \dots, f_m(x)); \quad h(x) = (h_1(x), \dots, h_p(x))$$

Assumption:  $p < n$

# Feasibility

A point  $x \in \mathcal{D}$  is *feasible* if  $f(x) \leq 0$  and  $h(x) = 0$ .

A feasible point is *strictly feasible* if  $f(x) < 0$ .

The set  $X$  of all feasible points is called the *feasible set*.

If  $X = \emptyset$ , then the optimization problem is *infeasible*. Otherwise it is *feasible*.

# Optimality

The *optimal value* of (1) is

$$p^* = \inf \{f_0(x) | f(x) \leq 0, h(x) = 0\}$$

If (1) is infeasible, then  $p^* = \infty$ .

If  $\exists x_k \in \mathcal{D}$ :  $\lim_{k \rightarrow \infty} f_0(x_k) = -\infty$ , then the optimization problem is *unbounded from below*, and  $p^* = -\infty$ .

$x^*$  is an *optimal point* or *solution* if  $x^*$  is feasible and  $f_0(x^*) = p^*$ .

The set of all optimal points is called the *optimal set*.

If the optimal set is nonempty we say that the optimal value is *attained* or *achieved*. Otherwise it is not attained or achieved.

A feasible point  $x$  for which  $f_0(x) \leq p^* + \epsilon$ , where  $\epsilon > 0$ , is called  $\epsilon$ -*suboptimal*.

# Local Optimality

A feasible point  $x$  is called *locally optimal* if  $\exists R > 0$ :

$$f_0(x) = \inf \{ f_0(z) \mid f(z) \leq 0, h(z) = 0, \|z - x\|_2 \leq R \}$$

A locally optimal point is not necessarily optimal. However, the converse is always true.

To distinguish a local optimal point from an optimal point we sometimes say that an optimal point is *globally optimal*.

# Feasibility Problem

When  $f_0(x) = 0 \ \forall x \in \mathcal{D}$ , then  $p^\star = 0$  if  $X \neq \emptyset$ , and  $p^\star = \infty$  if  $X = \emptyset$ .

Such an optimization problem is called a *feasibility problem* and solving it amounts to finding an  $x \in \mathcal{D}$  that satisfies the constraints.

# Equivalent Problems

Two optimization problems are said to be *equivalent* if the solution from one of them can be found from the other one with simple manipulations and vice versa.



# Convexity

A set  $C$  is called *convex* if  $\forall x, y \in C$  and  $\forall \theta \in [0, 1]$  it holds that  $\theta x + (1 - \theta)y \in C$ .

We say that a function  $f : \mathbf{R} \rightarrow \mathbf{R}$  is *convex* if its domain is convex and if  $\forall \theta \in [0, 1]$  it holds that

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

The function is called *strictly convex* if the inequality holds strictly whenever  $x \neq y$  and  $\theta \in (0, 1)$ .

We say that a function  $f$  is *concave* if  $-f$  is convex and similarly for strict concavity.

A vector valued function is convex if all its component functions are convex.

# The Epigraph and the Perspective Function

The *epigraph* of a function  $f$  is

$$\mathbf{epi} f = \{(x, t) \mid f(x) \leq t\}$$

A function  $f$  is convex if and only if  $\mathbf{epi} f$  is a convex set.

The *perspective* of a function  $f : \mathbf{R}^n \rightarrow \mathbf{R}$  is  $g : \mathbf{R}^n \times \mathbf{R} \rightarrow \mathbf{R}$  for which  $g(x, y) = yf(x/y)$  with domain  $\{(x, y) \mid y > 0, x/y \in \mathbf{dom} f\}$ .

If  $f$  is convex so is its perspective.

# Conditions for Convexity

Assume  $f$  differentiable. Then  $f$  is convex if and only if **dom**  $f$  convex and

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) \quad (2)$$

$\forall x, y \in \mathbf{dom} f$ .

Strict convexity holds if and only if the inequality holds strictly whenever  $x \neq y$ .

If  $f$  is twice differentiable, then  $f$  is convex if and only if **dom**  $f$  convex and

$$\nabla^2 f(x) \succeq 0$$

$\forall x \in \mathbf{dom} f$ .

If  $\nabla^2 f(x) \succ 0 \ \forall x \in \mathbf{dom} f$  then  $f$  is strictly convex. The converse is however not true.

# Convex Optimization Problem

The optimization problem in (1), i.e.

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i \in \mathbf{N}_m \\ & h_i(x) = 0, \quad i \in \mathbf{N}_p\end{array}$$

is a *convex optimization problem* if  $f_0$ , and  $f$  are convex functions, and if  $h(x)$  is an *affine* function, i.e. both convex and concave.

Convex optimization problems have convex feasible sets. The converse is however not true.

# Condition for Optimality

For the convex optimization problem (1) assume  $f_0$  differentiable.

Then  $x$  is optimal if and only if  $x \in X$  and

$$\nabla f_0(x)^T(y - x) \geq 0, \quad \forall y \in X \quad (3)$$

where  $X$  is the feasible set.

# Equality Constrained Optimization Problem

Consider

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & Ax = b\end{array}$$

where  $f_0 : \mathbf{R}^n \rightarrow \mathbf{R}$  differentiable and  $A \in \mathbf{R}^{p \times n}$   $b \in \mathbf{R}^p$ .

Define *Lagrangian*  $L : \mathbf{R}^n \times \mathbf{R}^p \rightarrow \mathbf{R}$  by

$$L(x, \mu) = f_0(x) + \mu^T h(x)$$

where  $h(x) = Ax - b$ . Then  $x$  is optimal if and only if there exist  $\mu$  such that  $x$  satisfies

$$\begin{aligned}\nabla_x L(x, \mu) &= 0 \\ h(x) &= 0\end{aligned}$$

# Lagrangian for General Non-Convex Problem

Consider (1), and define Lagrangian  $L : \mathbf{R}^n \times \mathbf{R}^m \times \mathbf{R}^p \rightarrow \mathbf{R}$  via

$$L(x, \lambda, \mu) = f_0(x) + \lambda^T f(x) + \mu^T h(x) \quad (4)$$

with domain  $\mathcal{D} \times \mathbf{R}^m \times \mathbf{R}^p$ .

Here  $\lambda$  and  $\mu$  are called the *Lagrange multipliers* or *dual variables*.

The *Lagrange dual function*  $g : \mathbf{R}^m \times \mathbf{R}^p \rightarrow \mathbf{R}$  is defined as

$$g(\lambda, \mu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \mu) \quad (5)$$

In case  $L$  is not bounded from below for some  $(\lambda, \mu)$ , it takes the value  $-\infty$ .

The function  $g$  is concave whether (1) is a convex optimization problem or not.

# Duality

For any  $\lambda \geq 0$  and any  $\mu$

$$g(\lambda, \mu) \leq p^*$$

We say that  $(\lambda, \mu)$  are *dual feasible* if  $\lambda \geq 0$  and they belong to the domain of  $g$ .



# The Conjugate Function

The *conjugate function*  $f^* : \mathbf{R}^n \rightarrow \mathbf{R}$  of a function  $f : \mathbf{R}^n \rightarrow \mathbf{R}$  is defined as

$$f^*(y) = \sup_{x \in \text{dom } f} \left( y^T x - f(x) \right)$$

# Application

Consider

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & Ax \leq b \\ & Cx = d\end{array}\tag{6}$$

where  $A \in \mathbf{R}^{m \times n}$ ,  $b \in \mathbf{R}^m$ ,  $C \in \mathbf{R}^{p \times n}$ , and  $d \in \mathbf{R}^p$ . Then

$$\begin{aligned}g(\lambda, \mu) &= \inf_{x \in \mathcal{D}} \left( f_0(x) + \lambda^T (Ax - b) + \mu^T (Cx - d) \right) \\ &= -\lambda^T b - \mu^T d + \inf_{x \in \mathcal{D}} \left( f_0(x) + (A^T \lambda + C^T \mu)^T x \right) \\ &= -\lambda^T b - \mu^T d - f_0^* \left( -A^T \lambda - C^T \mu \right)\end{aligned}$$

where the domain of  $g$  is  $\{(\lambda, \mu) \mid -A^T \lambda - C^T \mu \in \mathbf{dom} f_0^*\}$ .

# Lagrange Dual Problem

The *Lagrange dual problem* associated with (1) is defined as

$$\begin{array}{ll}\text{maximize} & g(\lambda, \mu) \\ \text{subject to} & \lambda \geq 0\end{array}\tag{7}$$

This provides a lower bound  $p^*$  of (1), called the *primal problem*.

For optimal value of the dual problem  $d^*$  it holds that  $d^* \leq p^*$ , called *weak duality*.

Optimal solution to (7) denoted  $(\lambda^*, \mu^*)$ , called *dual optimal* or *optimal Lagrange multipliers*.

Lagrange dual problem is convex optimization problem.

# Stopping Criteria

For any dual feasible  $(\lambda, \mu)$  and primal feasible  $x$  it holds that  $g(\lambda, \mu) \leq p^*$  implies that

$$f_0(x) - p^* \leq f_0(x) - g(\lambda, \mu)$$

and hence  $x$  is  $\epsilon$ -suboptimal with  $\epsilon = f_0(x) - g(\lambda, \mu)$ .

Similarly  $(\lambda, \mu)$  is  $\epsilon$ -suboptimal for the dual problem.

We call  $\epsilon$  the *duality gap*. For any dual feasible  $(\lambda, \mu)$  and primal feasible  $x$  it holds that

$$p^* \in [g(\lambda, \mu), f_0(x)]; \quad d^* \in [g(\lambda, \mu), f_0(x)]$$

# Strong Duality

The difference  $p^* - d^*$  called the *optimal duality gap*.

If  $d^* = p^*$ , we then say that *strong duality* holds—not true in general.

Often need *constraint qualifications* and convexity for strong duality.

*Slater's condition*: There exist  $x \in \mathbf{relint} \mathcal{D} \cap X$  such that  $f(x) < 0$ , i.e. *strict feasibility*.

Slater's condition also implies that the dual optimal value is attained when  $d^* > -\infty$ , i.e. there exist  $(\lambda^*, \mu^*)$  such that  $g(\lambda^*, \mu^*) = d^*$ .

# Karush-Kuhn-Tucker (KKT) conditions

Consider the primal problem in (1) and the corresponding dual problem in (7) and assume that  $f_0$ ,  $f$  and  $h$  are differentiable. Given primal and dual optimal points  $x^*$  and  $(\lambda^*, \mu^*)$  with zero duality gap it holds that

$$\nabla L(x^*, \lambda^*, \mu^*) = 0 \quad (8)$$

$$f(x^*) \leq 0 \quad (9)$$

$$h(x^*) = 0 \quad (10)$$

$$\lambda^* \geq 0 \quad (11)$$

$$\lambda_i^* f_i(x^*) = 0, \quad i \in \mathbf{N}_m \quad (12)$$

The above conditions are also sufficient if the primal problem is convex.

# Necessary Conditions for Nonconvex Problems

For non-convex problems there rarely exist primal and dual optimal points  $x^*$  and  $(\lambda^*, \mu^*)$  with zero duality gap.

Consider (1) where we assume that  $f_0$ ,  $f$  and  $h$  are differentiable.

Let  $x^*$  be a local optimal point and let  $\mathcal{A} \subset \mathbf{N}_m$  be the set of  $i$  such that  $f_i(x^*) = 0$ , i.e. the set of *active inequality constraints*. If  $\nabla f_i(x^*)$ ,  $i \in \mathcal{A}$  and  $\nabla h_i(x^*)$ ,  $i \in \mathbf{N}_p$  are linearly independent, then there exist  $\lambda \in \mathbf{R}^m$  and  $\mu \in \mathbf{R}^p$  such that

$$\nabla L(x^*, \lambda, \mu) = 0$$

$$f(x^*) \leq 0$$

$$h(x^*) = 0$$

$$\lambda \geq 0$$

$$\lambda_i f_i(x^*) = 0, \quad i \in \mathbf{N}_m$$

# Sufficient Conditions for Nonconvex Problems

Let

$$M = \{y \in \mathbf{R}^n \mid \nabla h_i(x^*)y = 0, \nabla f_j(x^*)y = 0, \forall i \in \mathbf{N}_p, \forall j \in J\}$$

where

$$J = \{j \in \mathbf{N}_m : f_j(x^*) = 0, \lambda_j > 0\}$$

Assume that  $f_0, f_i, i \in \mathbf{N}_m$  and  $h_j, j \in \mathbf{N}_p$  are twice continuously differentiable.

Sufficient condition for  $x^*$  to be a local optimal point is  $\exists(\lambda, \mu)$ :

$$\nabla L(x^*, \lambda, \mu) = 0$$

$$f(x^*) \leq 0$$

$$h(x^*) = 0$$

$$\lambda \geq 0$$

$$\lambda_i f_i(x^*) = 0, \quad i \in \mathbf{N}_m$$

and that  $y \nabla^2 L(x^*, \lambda, \mu) y^T > 0$  for all  $y \in M$ .



# Entropy

Let

$$\mathcal{D}_n = \left\{ p = (p_1, \dots, p_n) \in [0, 1]^n \mid p_k \geq 0, k \in \mathbf{N}_n; \sum_{k=1}^n p_k = 1 \right\}$$

be the set of finite probability distributions with  $n$  events.

The entropy  $H_n : \mathcal{D}_n \rightarrow \mathbf{R}$  should satisfy

1.  $H_n(p) = H_n(\pi(p))$ , where  $\pi$  is any permutation function.
2.  $H_n(p) \leq H_n(\bar{p})$ , where  $\bar{p} = (1/n, \dots, 1/n)$ .
3.  $H_n(p) = H_{n+1}(q)$ , where  $q = (p, 0)$
4. If  $r \in \mathcal{D}_{mn}$  is a joint probability distribution with marginal distributions  $p \in \mathcal{D}_n$  and  $q \in \mathcal{D}_n$ , then

$$H_{mn}(r) = H_n(p) + \sum_{k: p_k \neq 0} p_k H_m(r_k/p_k), \text{ where } r_k = (r_{k,1}, \dots, r_{k,m}).$$

$\implies$

$$H_n(p) = -k \sum_{k=1}^n p_k \log p_k$$

# Maximum Entropy Principle

Assume that we observe values of a random variable and estimate the expectation of it. What is then the most likely distribution of the random variable? The *maximum entropy principle* says that it is the one that maximizes the entropy among all possible probability distributions that have the same mean.

$$\text{maximize}_p \quad - \sum_{k=1}^n p_k \ln p_k \quad (13)$$

$$\text{subject to } Ap = b \quad (14)$$

$$\mathbf{1}^T p = 1 \quad (15)$$

$$p \geq 0 \quad (16)$$

where  $A \in \mathbf{R}^{m \times n}$ .

*Convex Optimization Problem*

# Expectation Constraints

Every row in the first constraint is an expectation constraint.

Define random variables  $a_i$  that takes values  $A_{i,j}$  with probability  $p_j$ . Then the expected value of  $a_i$  is  $\sum_{j=1}^n A_{i,j}p_j$ , which is the left hand side of the  $i$ th row.

In an application the right hand side  $b_i$  could be obtained from an empirical estimate of the expected value.

Simple example of  $a_i$  is when  $A_{i,j} = 1$  if  $j = i$  and zero otherwise. Hence the expected value is  $p_j$  and if the right hand side is an empirical estimate of this expected value, we have effectively constrained  $p_j$  to be equal to its empirical estimate.

If  $a_i$  takes values  $f_j$  with probability  $p_j$  then by defining  $A_{i,j} = f_j^2$  we define a constraint on the second moment of the random variable.

# Example of Maximum Entropy Problem

Consider

$$\text{maximize}_p H_n(p) \quad (17)$$

$$\text{subject to } f^T p = b \quad (18)$$

$$\mathbf{1}^T p = 1 \quad (19)$$

$$p \geq 0 \quad (20)$$

where  $f \in \mathbf{R}^n$  and  $b \in \mathbf{R}$  given.

Lagrangian  $L : \mathbf{R}^n \times \mathbf{R} \times \mathbf{R} \rightarrow \mathbf{R}$  where

$$L(p, \lambda, \mu) = H_n(p) + \lambda(b - f^T p) + \mu(1 - \mathbf{1}^T p)$$

is a concave function.

# Lagrange Dual Function

Maximum of Lagrange function subject to  $p_k \geq 0$  obtained when

$$\frac{\partial L}{\partial p_k} = -\ln p_k - 1 - \lambda f_k - \mu = 0$$

$\implies$

$$p_k = \frac{e^{-\lambda f_k}}{e^{1+\mu}} \geq 0$$

Lagrange dual function  $g : \mathbf{R} \times \mathbf{R} \rightarrow \mathbf{R}$  where

$$g(\lambda, \mu) = \frac{\sum_{k=1}^n e^{-\lambda f_k}}{e^{1+\mu}} + \lambda b + \mu$$

which is a convex function.

# Minimization of Lagrange Dual Function

Minimum of  $g$  with respect to  $\mu$  when

$$\frac{\partial g}{\partial \mu} = -\frac{\sum_{k=1}^n e^{-\lambda f_k}}{e^{1+\mu}} + 1 = 0$$

$\implies$

$$\Phi(\lambda) := e^{1+\mu} = \sum_{k=1}^n e^{-\lambda f_k}$$

where  $\phi : \mathbf{R} \rightarrow \mathbf{R}_{++}$ .

Back substitution into the Lagrange dual function results in  $h : \mathbf{R} \rightarrow \mathbf{R}$  defined by

$$h(\lambda) = \lambda b + \ln \Phi(\lambda)$$

# Minimization of $h$

Minimum of  $h$  with respect to  $\lambda$  when

$$\frac{\partial h}{\partial \lambda} = b + \frac{\partial \Phi}{\partial \lambda} / \Phi(\lambda) = b - \frac{\sum_{k=1}^n f_k e^{-\lambda f_k}}{\Phi(\lambda)} = 0$$

$$\Longleftrightarrow$$

$$\sum_{k=1}^n (f_k - b) e^{-\lambda f_k} = 0$$

This equation has a solution  $\lambda$  if  $f_k \neq f_l$  for some  $k \neq l$ , since then  $G(\lambda) = \sum_{k=1}^n (f_k - b) e^{-\lambda f_k}$  is strictly decreasing as function of  $\lambda$  with

$$\lim_{\lambda \rightarrow -\infty} G(\lambda) = \infty, \quad \lim_{\lambda \rightarrow +\infty} G(\lambda) = -\infty$$

# Natural Parameters

If we do not want to parameterize the distribution in terms of  $b$ , then we can do it in terms of the *natural parameter*  $\lambda$ , i.e.

$$p_k = \frac{e^{-\lambda f_k}}{\sum_{l=1}^n e^{-\lambda f_l}}$$

Distribution is called the *Categorical*, *Gibbs* or *Boltzmann* distribution.

Belongs to the family of *Exponential distributions*

Normalization used in logistic regression:

$$p_k = \begin{cases} \frac{e^{z_k}}{1 + \sum_{l=1}^{n-1} e^{z_l}}, & k \in \mathbf{N}_{n-1} \\ \frac{1}{1 + \sum_{l=1}^{n-1} e^{z_l}}, & k = n \end{cases}$$

where  $z_k = \lambda(f_n - f_k)$ ,  $k \in \mathbf{N}_{n-1}$ .



# Binary-Valued Random Vectors

Introduce the bijection  $\mathcal{D}_x = \{0, 1\}^m \leftrightarrow \mathbf{N}_n$ , where  $n = 2^m$  such that for  $x \in \mathcal{D}_x$  and  $k \in \mathbf{N}_n$  it holds that  $k = \sum_{l=1}^n x_l 2^{l-1}$ , where  $x = (x_1, \dots, x_m)$  with  $x_l \in \{0, 1\}$ .

With abuse of notation we let  $p(x) = p_k$ , with  $p = (p_1, \dots, p_n) \in \mathcal{D}_n$ .

Then  $H_n(p) = -\sum_{x \in \mathcal{D}_x} p(x) \ln(p(x))$ .

# Maximum Entropy for Binary-Valued Random Vectors

Constraints on the first moment of  $x$  and on the cross-moments between the components of  $x$ :

$$\text{maximize}_p H_n(p) \quad (21)$$

$$\text{subject to } \sum_{x \in \mathcal{D}_x} xp(x) = m \quad (22)$$

$$\sum_{x \in \mathcal{D}_x} xx^T p(x) = M \quad (23)$$

$$\sum_{x \in \mathcal{D}_x} p(x) = 1 \quad (24)$$

$$p(x) \geq 0, \quad x \in \mathcal{D}_x \quad (25)$$

where with abuse of notation  $m$  is the vector of first moments, and  $M$  is the matrix of second moments, where we do not specify the diagonal.

# Lagrangian

Lagrangian  $L : \mathbf{R}^n \times \mathbf{R}^m \times \mathbf{S}^m \times \mathbf{R} \rightarrow \mathbf{R}$  via

$$L(p, \lambda, \Lambda, \mu) = H_n(p) + \lambda^T \left( \sum_{x \in \mathcal{D}_x} xp(x) - m \right) \\ + \text{tr} \Lambda \left( \sum_{x \in \mathcal{D}_x} xx^T p(x) - M \right) + \mu \left( \sum_{x \in \mathcal{D}_x} p(x) - 1 \right)$$

where  $\Lambda$  has a zero diagonal.

Maximized when

$$\frac{\partial L}{\partial p(x)} = -\ln p(x) - 1 + \lambda^T x + \text{tr} \Lambda xx^T + \mu = 0$$

# Ising Distribution

$$p(x) = \exp \left( \lambda^T x + \text{tr} \Lambda x x^T - 1 + \mu \right)$$

Lagrange dual function  $g : \mathbf{R}^m \times \mathbf{S}^m \times \mathbf{R} \rightarrow \mathbf{R}$  by back substitution of  $p$  into the Lagrangian. Minimizing this function with respect to  $\mu$  results in choosing  $\mu$  such that the probabilities sum up to one. With  $A = 1 - \mu$  this holds if

$$A = \ln \sum_{x \in \mathcal{D}_x} \exp \left( \lambda^T x + \text{tr} \Lambda x x^T \right)$$

We see that  $A$  is a function of  $\lambda$  and  $\Lambda$ , and we write  $A : \mathbf{R}^m \times \mathbf{S}^m \rightarrow \mathbf{R}$ .

# Equivalent Formulation

With the *energy function*  $E : \{0, 1\}^m \rightarrow \mathbf{R}$  given by

$$E(x) = -\lambda^T x - \mathbf{tr} \Lambda x x^T$$

where  $\Lambda$  has a zero diagonal, we have

$$p(x) = \exp(-E(x) - A(\lambda, \Lambda))$$

where

$$A(\lambda, \Lambda) = \ln \sum_x \exp(-E(x))$$

## Relation to Moments

Back-substitute  $\mu = 1 - A(\lambda, \Lambda)$  into the Lagrange dual function and obtain  $h : \mathbf{R}^m \times \mathbf{S}^m \rightarrow \mathbf{R}$  given by

$$h(\lambda, \Lambda) = A(\lambda, \Lambda) - \lambda^T m - \text{tr } \Lambda M$$

Minimum of  $h$  with respect to  $(\lambda, \Lambda)$  when

$$\begin{aligned} \frac{\partial h}{\partial \lambda} &= \frac{\partial A}{\partial \lambda} - m = 0 \\ \frac{\partial h}{\partial \Lambda} &= \frac{\partial A}{\partial \Lambda} - M = 0 \end{aligned}$$

# Partial Derivatives

$$\begin{aligned}\frac{\partial A}{\partial \lambda} &= \frac{\sum_{x \in \mathcal{D}_x} x \exp(-E(x))}{\sum_{x \in \mathcal{D}_x} \exp(-E(x))} = \frac{\sum_{x \in \mathcal{D}_x} x \exp(-E(x) - A(\lambda, \Lambda))}{\sum_{x \in \mathcal{D}_x} \exp(-E(x) - A(\lambda, \Lambda))} \\ &= \sum_{x \in \mathcal{D}_x} x p(x) \\ \frac{\partial A}{\partial \Lambda} &= \frac{\sum_{x \in \mathcal{D}_x} x x^T \exp(-E(x))}{\sum_{x \in \mathcal{D}_x} \exp(-E(x))} = \frac{\sum_{x \in \mathcal{D}_x} x x^T \exp(-E(x) - A(\lambda, \Lambda))}{\sum_{x \in \mathcal{D}_x} \exp(-E(x) - A(\lambda, \Lambda))} \\ &= \sum_{x \in \mathcal{D}_x} x x^T p(x)\end{aligned}$$

We should match the moments!

To solve the above equations with respect to  $(\lambda, \Lambda)$  is not easy in general, and specifically not when the dimension of  $x$  is large.

# Entropy for Infinite Probability Distributions

For infinite probability distribution  $p : \mathbf{R} \rightarrow \mathbf{R}_+$  define entropy as

$$H(p) = - \int_{-\infty}^{+\infty} p(x) \log p(x) dx$$



# Normal Distribution

Let  $\mathcal{D}$  be the space of real-valued functions defined on  $\mathbf{R}^n$  and consider  $p \in \mathcal{D}$  subject to  $p(x) \geq 0$ ,  $\int_{\mathbf{R}^n} p(x) dx = 1$ ,  $\int_{\mathbf{R}^n} x p(x) dx = m$  and  $\int_{\mathbf{R}^n} x x^T p(x) dx = M$ , where  $m \in \mathbf{R}^n$  and  $M \in \mathbf{S}_+$  are the first and second moments of the distribution, respectively. The maximum entropy problem is

$$\text{maximize}_{p \in \mathcal{D}} H(p) \quad (26)$$

$$\text{subject to } \int_{\mathbf{R}^n} x p(x) dx = m \quad (27)$$

$$\int_{\mathbf{R}^n} x x^T p(x) dx = M \quad (28)$$

$$\int_{\mathbf{R}^n} p(x) dx = 1 \quad (29)$$

$$p(x) \geq 0, \quad x \in \mathbf{R}^n \quad (30)$$

## Lagrangian

Lagrangian  $L : \mathcal{D} \times \mathbf{R} \times \mathbf{S}^n \times \mathbf{R} \rightarrow \mathbf{R}$  with

$$L(p, \lambda, \Lambda, \mu) = - \int_{\mathbf{R}^n} p(x) \ln p(x) dx + \lambda^T \left( m - \int_{\mathbf{R}^n} x p(x) dx \right) \quad (31)$$

$$+ \frac{1}{2} \text{tr} \left( \Lambda \left( M - \int_{\mathbf{R}^n} x x^T p(x) dx \right) \right) \quad (32)$$

$$+ \mu \left( 1 - \int_{\mathbf{R}^n} p(x) dx \right) \quad (33)$$

First variation of the Lagrangian:

$$\delta L = - \int_{\mathbf{R}^n} \delta p \left[ \ln p + 1 + \lambda^T x + \frac{1}{2} x^T \Lambda x + \mu \right] dx$$

which should be non-positive for all  $\delta p$  when  $p$  is optimal. Hence expression in bracket must vanish and optimal distribution is

$$p(x) = e^{-1-\mu-\lambda^T x - \frac{1}{2} x^T \Lambda x} \geq 0$$

# Lagrange Dual Function

Lagrange dual function  $g : \mathbf{R} \times \mathbf{S}^n \times \mathbf{R} \rightarrow \mathbf{R}$  defined by

$$g(\lambda, \Lambda, \mu) = \int_{\mathbf{R}^n} e^{-1-\mu-\lambda^T x - \frac{1}{2}x^T \Lambda x} dx + \lambda^T m + \frac{1}{2} \text{tr} \Lambda M + \mu \quad (34)$$

$$= \int_{\mathbf{R}^n} e^{-1-\mu+\frac{1}{2}\lambda^T \Lambda^{-1} \lambda - \frac{1}{2}(x+\Lambda^{-1}\lambda)^T \Lambda (x+\Lambda^{-1}\lambda)} dx \quad (35)$$

$$+ \frac{1}{2} \lambda^T m + \text{tr} \Lambda M + \mu \quad (36)$$

and we determine the  $\mu$  that minimizes it by setting the partial derivative of  $g$  with respect to  $\mu$  equal to zero, which is equivalent to that  $p(x)$  integrates to one.

# Normalization

$$\int_{\mathbf{R}^n} p(x) dx = \int_{\mathbf{R}^n} e^{-1-\mu+\frac{1}{2}\lambda^T \Lambda^{-1} \lambda - \frac{1}{2}(x+\Lambda^{-1}\lambda)^T \Lambda (x+\Lambda^{-1}\lambda)} dx \quad (37)$$

$$= \frac{2^{n/2} e^{-1-\mu+\frac{1}{2}\lambda^T \Lambda^{-1} \lambda}}{\det \Lambda^{1/2}} \int_{\mathbf{R}} e^{-\bar{x}_1^2} d\bar{x}_1 \times \cdots \times \int_{\mathbf{R}} e^{-\bar{x}_n^2} d\bar{x}_n \quad (38)$$

$$= \frac{e^{-1-\mu+\frac{1}{2}\lambda^T \Lambda^{-1} \lambda} (2\pi)^{n/2}}{\sqrt{\det \Lambda}} = 1 \quad (39)$$

Hence  $\mu = \frac{1}{2}\lambda^T \Lambda^{-1} \lambda - 1 - \frac{1}{2} \ln \frac{\det \Lambda}{(2\pi)^n}$  and we get

$$p(x) = \sqrt{\frac{\det \Lambda}{(2\pi)^n}} e^{-\frac{1}{2}(x+\Lambda^{-1}\lambda)^T \Lambda (x+\Lambda^{-1}\lambda)}$$

# Back Substitution

Optimal  $\mu$  into the Lagrange dual function defines

$h : \mathbf{R}^n \times \mathbf{S}^n \rightarrow \mathbf{R}$  via

$$h(\lambda, \Lambda) = g \left( \lambda, \Lambda, \frac{1}{2} \lambda^T \Lambda^{-1} \lambda - 1 - \frac{1}{2} \ln \frac{\det \Lambda}{(2\pi)^n} \right) \quad (40)$$

$$= \frac{1}{2} \lambda^T \Lambda^{-1} \lambda - \frac{1}{2} \ln \frac{\det \Lambda}{(2\pi)^n} + \lambda^T m + \frac{1}{2} \text{tr} \Lambda M \quad (41)$$

Minimizing  $h$  from:

$$\frac{\partial h}{\partial \lambda} = \Lambda^{-1} \lambda + m = 0 \quad (42)$$

$$\frac{\partial h}{\partial \Lambda} = -\frac{1}{2} \Lambda^{-1} \lambda \lambda^T \Lambda^{-1} - \frac{1}{2} \Lambda^{-1} + \frac{1}{2} M = -\frac{1}{2} m m^T - \frac{1}{2} \Lambda^{-1} + \frac{1}{2} M = 0 \quad (43)$$

# Non-Natural Parameters

With  $\Sigma = M - mm^T$  it holds that  $\Lambda = \Sigma^{-1}$  and we have  $\lambda = -\Sigma^{-1}m$  and

$$p(x) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} e^{-\frac{1}{2}(x-m)^T \Sigma^{-1}(x-m)}$$

In case  $h$  is expressed in terms of  $(m, \Sigma)$  instead of in terms of  $(\lambda, \Lambda)$  it will not be a convex function.

**Natural parameters make convex!**

# Prediction

For a joint distribution  $r : \mathbf{R}^m \times \mathbf{R}^n \rightarrow \mathbf{R}_+$  of two random variables with marginal distributions  $p : \mathbf{R}^m \rightarrow \mathbf{R}_+$  and  $q : \mathbf{R}^n \rightarrow \mathbf{R}_+$  we are given an observation  $x$  from the distribution  $p$  and would like to *predict* a value  $y$  from the distribution  $q$ .

The conditional distribution of  $y$  given  $x$ , i.e.  $f : \mathbf{R}^m \times \mathbf{R}^n \rightarrow \mathbf{R}_+$  defined as

$$f(x, y) = \frac{r(x, y)}{p(x)}$$

is one solution. We could also maximize it w.r.t.  $y$  to get a point estimate.

Another possibility is to look for a predictor  $g : \mathbf{R}^m \rightarrow \mathbf{R}^n$  that solves

$$\text{minimize}_g E_r (y - g(x))^T (y - g(x))$$

# Conditional Expectation

$$E_r (y - g(x))^T (y - g(x)) \quad (44)$$

$$= \int_{\mathbf{R}^m} p(x) \int_{\mathbf{R}^n} f(x, y) \left( y^T y - 2g^T(x)y + g^T(x)g(x) \right) dy dx \quad (45)$$

$$= \int_{\mathbf{R}^m} p(x) \left( E_f y^T y - (E_f y)^T E_f y + (g(x) - E_f y)^T (g(x) - E_f y) \right) dx \quad (46)$$

Minimum attained for  $g(x) = E_f y$ , i.e. the conditional expectation of  $y$  given  $x$ .

Minimal value of objective function is covariance of  $y$  conditioned on  $x$ .



# Normal Distribution Case

Let

$$r(z) = \frac{1}{\sqrt{(2\pi)^{m+n} \det \Sigma}} e^{-\frac{1}{2}(z-\mu)^T \Sigma^{-1}(z-\mu)}$$

where  $z = (x, y)$ ,  $\mu = (\mu_x, \mu_y)$  and where

$$\Sigma = \begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{xy}^T & \Sigma_y \end{bmatrix}$$

The *Schur complement formula*:

$$\begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{xy}^T & \Sigma_y \end{bmatrix} = \begin{bmatrix} I & 0 \\ \Sigma_{xy}^T \Sigma_x^{-1} & I \end{bmatrix} \begin{bmatrix} \Sigma_x & 0 \\ 0 & \Sigma_y - \Sigma_{xy}^T \Sigma_x^{-1} \Sigma_{xy} \end{bmatrix} \begin{bmatrix} I & \Sigma_x^{-1} \Sigma_{xy} \\ 0 & I \end{bmatrix}$$

where

$$\begin{bmatrix} I & \Sigma_x^{-1} \Sigma_{xy} \\ 0 & I \end{bmatrix}^{-1} = \begin{bmatrix} I & -\Sigma_x^{-1} \Sigma_{xy} \\ 0 & I \end{bmatrix}$$

# Factorization of Distribution

We can write  $r(z) = p(x)f(x, y)$ , where

$$p(x) = \frac{1}{\sqrt{(2\pi)^m \det \Sigma_x}} e^{-\frac{1}{2}(x-\mu_x)^T \Sigma_x^{-1}(x-\mu_x)}$$

and where

$$f(x, y) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma_{y|x}}} e^{-\frac{1}{2}(y-\mu_{y|x})^T \Sigma_{y|x}^{-1}(y-\mu_{y|x})}$$

where

$$\mu_{y|x} = \mu_y + \Sigma_{xy}^T \Sigma_x^{-1}(x - \mu_x); \quad \Sigma_{y|x} = \Sigma_y - \Sigma_{xy}^T \Sigma_x^{-1} \Sigma_{xy}$$

Hence  $f(x, y)$  is conditional distribution and  $\mu_{y|x}$  conditional expectation, which is an affine function in  $x$ .

# Affine Predictor

Consider general distribution  $r$  that is not necessarily normal and let  $g(x) = Ax + b$  with  $A \in \mathbf{R}^{m \times n}$  and  $b \in \mathbf{R}^m$ .

Let  $m_x = E_r x$ ,  $m_y = E_r y$  and  $\bar{x} = x - m_x$  and  $\bar{y} = y - m_y$  and define  $J : \mathbf{R}^{m \times n} \times \mathbf{R}^m \rightarrow \mathbf{R}_+$  as

$$J(A, b) = \frac{1}{2} E_r (y - Ax - b)^T (y - Ax - b) \quad (47)$$

$$= \frac{1}{2} \text{tr} E_r (y - Ax - b) (y - Ax - b)^T \quad (48)$$

$$= \frac{1}{2} \text{tr} E (\bar{y} - A\bar{x} + m_y - Am_x - b) (\bar{y} - A\bar{x} + m_y - Am_x - b)^T \quad (49)$$

$$= \frac{1}{2} \text{tr} E (\bar{y} - A\bar{x}) (\bar{y} - A\bar{x})^T \quad (50)$$

$$+ \frac{1}{2} (m_y - Am_x - b)^T (m_y - Am_x - b) \quad (51)$$

where we have used  $E\bar{x} = 0$  and  $E\bar{y} = 0$ , and where  $E$  denotes expectation with respect to the shifted distribution.

## Optimal Predictor

With  $D_x = E\bar{x}\bar{x}^T$  and  $D_{xy} = E\bar{x}\bar{y}^T$  optimal  $A$  and  $b$  satisfy

$$\frac{\partial J}{\partial b} = -m_y + Am_x + b = 0 \quad (52)$$

$$\frac{\partial J}{\partial A} = -D_{xy}^T + AD_x - m_y m_x^T + b m_x^T + A m_x m_x^T = 0 \quad (53)$$

with solution  $A = D_{xy}^T D_x^{-1}$  and  $b = m_y - D_{xy}^T D_x^{-1} m_x$ .

Optimal predictor:

$$Ax + b = m_y + D_{xy}^T D_x^{-1} (x - m_x)$$

which is in agreement with the normal distribution case.

The minimal value of  $J$  is  $\text{tr}(D_y - D_{xy}^T D_x^{-1} D_{xy})$ . In general larger than the covariance of  $y$  conditioned on  $x$ .

# Graphical Models

Consider a graph  $G = (V, E)$ , where  $V = \mathbf{N}_n$  is the set of vertices and where  $E \subset V \times V$  is a set of undirected edges connecting vertices in  $V$ .

We let the elements of  $V$  index components of an  $n$ -dimensional random variable.

We define similarly as before a distribution function by maximizing entropy under moment constraints. However, we only specify second order moments for combinations of components belonging to  $E$ .

# Graphical Ising Distribution

When we do this for an  $n$ -dimensional Ising distribution we obtain

$$p(x) = \exp \left( \sum_{k \in \mathbf{N}_n} \lambda_k x_k + \sum_{(i,j) \in E} \Lambda_{i,j} x_i x_j - A \right)$$

where

$$A(\lambda, \Lambda) = \ln \sum_{x \in \mathcal{D}_x} \exp \left( \sum_{k \in \mathbf{N}_n} \lambda_k x_k + \sum_{(i,j) \in E} \Lambda_{i,j} x_i x_j \right)$$

We notice that we just do not specify all the entries of neither  $\Lambda$  nor  $M$ .

## Matching conditions

If we let  $\Lambda_{i,j} = 0$  for  $(i,j) \notin E$ , then we may write the distribution also in this case using  $\text{tr } \Lambda x x^T$ . The Lagrange dual function will therefore look the same and so will the optimality conditions, except for the fact that some of them are void, i.e. we only have

$$\begin{aligned}\frac{\partial h}{\partial \lambda} &= \frac{\partial A}{\partial \lambda} - m = 0 \\ \frac{\partial h}{\partial \Lambda_{i,j}} &= \frac{\partial A}{\partial \Lambda_{i,j}} - M_{i,j} = 0, \quad (i,j) \in E\end{aligned}$$

Also for this case it is in general difficult to solve the optimality conditions.

## Graphical Normal Distribution

Similarly, when we derive the normal distribution the only difference is that  $\Lambda$  is such that  $\Lambda_{i,j} = 0$  for  $(i,j) \notin E$ . We do not need to specify  $M_{i,j}$  for  $(i,j) \notin E$ .

The Lagrange dual function will look the same. We first minimize it with respect to  $\lambda$ , which results in  $\lambda = -\Lambda m$ . We then back substitute in order to obtain the function  $f : \mathbf{S}^n \rightarrow \mathbf{R}$  given by

$$f(\Lambda) = -\frac{1}{2} \ln \frac{\det \Lambda}{(2\pi)^n} + \frac{1}{2} \text{tr} \Lambda \Sigma$$

where  $\Sigma = M - mm^T$ .

We need to minimize  $f$  subject to the constraint that  $\Lambda_{i,j} = 0$  for  $(i,j) \notin E$  in order to express  $\Lambda$  in terms of the moments.

This is a convex optimization problem with a linear constraint on  $\Lambda$ .



## Further Analysis

Lagrangian  $\mathcal{L} : \mathbf{S}^n \times \mathbf{S}^n$  defined by

$$\mathcal{L}(\Lambda, \Gamma) = -\frac{1}{2} \ln \frac{\det \Lambda}{(2\pi)^n} + \frac{1}{2} \text{tr} \Lambda \Sigma + \frac{1}{2} \text{tr} \Gamma \Lambda$$

where  $\Gamma_{i,j} = 0$  for  $(i,j) \in E$ .

Optimality conditions

$$\frac{\partial \mathcal{L}}{\partial \Lambda} = -\frac{1}{2} \Lambda^{-1} + \frac{1}{2} \Sigma + \frac{1}{2} \Gamma = 0$$

together with  $\Lambda_{i,j} = 0$  for  $(i,j) \notin E$  and  $\Gamma_{i,j} = 0$  for  $(i,j) \in E$ .

# Positive Definite Completion

We realize that we have full freedom in selecting the entries of  $\Sigma + \Gamma$  for indexes  $(i, j) \notin E$  by choosing  $\Gamma$ . Hence it does not matter that we did not specify these entries for  $M$ . We may take them equal to any value. Because of this we let  $\Sigma_{i,j} = 0$  for  $(i, j) \notin E$ .

However,  $\Sigma + \Gamma$  will be the covariance matrix of the normal distribution, and hence has to be positive definite.

In summary we want to find  $\Gamma$  to complete the covariance matrix in such a way that it is positive definite and such that its inverse has zeros for entries not in the set  $E$ .

This is called the *positive definite completion* of  $\Sigma$ .

# Conditional Independence

Consider

$$p(x) = \sqrt{\frac{\det \Lambda}{\pi^n}} e^{-(x-m)^T \Lambda (x-m)}$$

Let  $A$ ,  $B$ , and  $C$  be three mutually disjoint subsets of  $V$  and such that if we remove the vertices indexed by  $C$ , then there are no paths connecting the vertices in  $A$  with the vertices in  $B$ .

Also assume  $V = A \cup B \cup C$ .

Order such that  $x = (x_A, x_B, x_C)$ .

$$\begin{aligned} &\implies \\ \Lambda &= \begin{bmatrix} * & 0 & * \\ 0 & * & * \\ * & * & * \end{bmatrix} \end{aligned}$$

and

$$p(x) = p_{A,B|C}(x) p_C(x_C) = p_{A|C}(x_A, x_C) p_{B|C}(x_B, x_C) p_C(x_C)$$

# Markov Random Field

In general  $p$  can be factorized as

$$p(x) = \prod_{C \in \mathcal{C}} f_C(x_C)$$

for some functions  $f_C : \mathbf{R}^{|C|} \rightarrow \mathbf{R}_+$ , where  $\mathcal{C}$  is the set of all *cliques* of  $G$ , i.e. the set of all complete subgraphs of  $G$ .

The above Markov property holds for general graphical models defined on undirected graphs, and specifically also for the Ising model.

# Maximum Likelihood Estimation

For a finite distribution  $p \in \mathcal{D}_n$  that depends on a parameter  $\lambda \in \mathbf{R}$  we may define the likelihood function  $\ell : \mathbf{R}^N \times \mathbf{R} \rightarrow [0, 1]$  based on  $N$  samples  $f_{k_i}$ ,  $i \in \mathbf{N}_N$  of the random variable as

$$\ell(f_{k_1}, \dots, f_{k_N}; \lambda) = \prod_{i=1}^N p_{k_i}(\lambda)$$

Then the estimate of  $\lambda$  is obtained by maximizing  $\ell$ .

For Categorical distribution:

$$\ln \ell(f_{k_1}, \dots, f_{k_N}; \lambda) = - \sum_{i=1}^N \lambda f_{k_i} - N \ln \Phi(\lambda)$$

With  $b = \frac{1}{N} \sum_{i=1}^N f_{k_i}$  minimizing "Lagrange dual function"  $h(\lambda)$  is equivalent to maximizing  $\ln \ell(f_{k_1}, \dots, f_{k_N}; \lambda)$ .

# Ising Distribution

For Ising distribution  $\ell : \{0, 1\}^{mN} \times \mathbf{R}^m \times \mathbf{S}^m \rightarrow [0, 1]$  based on  $N$  samples  $x_i \in \{0, 1\}^m$ ,  $i \in \mathbf{N}_N$ :

$$\ell(x_1, \dots, x_N; \lambda, \Lambda) = \prod_{i=1}^N p(x_i)$$

we have

$$\ln \ell(x_1, \dots, x_N; \lambda, \Lambda) = \lambda^T \sum_{i=1}^N x_i + \text{tr} \Lambda \sum_{i=1}^N x_i x_i^T - N A(\lambda, \Lambda)$$

With  $m = \frac{1}{N} \sum_{i=1}^N x_i$  and  $M = \frac{1}{N} \sum_{i=1}^N x_i x_i^T$  minimizing "Lagrange Dual function"  $h(\lambda, \Lambda)$  is equivalent to maximizing the likelihood function.

Also true for graphical Ising model.

## Normal Distribution

For the normal distribution with  $\theta = (\lambda, \Lambda) \in \mathbf{R}^n \times \mathbf{S}^n$ :

$$\ln \ell(x_1, \dots, x_N; \theta) = -\frac{1}{2} \text{tr} \Lambda \left( \sum_{i=1}^N x_i x_i^T \right) - \lambda^T \sum_{i=1}^N x_i - \frac{N}{2} \lambda^T \Lambda^{-1} \lambda \quad (54)$$

$$+ \frac{N}{2} \ln \frac{\det \Lambda}{(2\pi)^n} \quad (55)$$

With

$$m = \frac{1}{N} \sum_{i=1}^N x_i; \quad M = \frac{1}{N} \sum_{i=1}^N x_i x_i^T$$

minimizing "Lagrange dual function"  $h(\Lambda)$  is equivalent to maximizing the likelihood function.

The solution has already been computed and with  $\Sigma = M - mm^T$  we have  $\Lambda = \Sigma^{-1}$  and  $\lambda = -\Sigma^{-1}m$ .

Similar results for graphical normal distribution

## Generalizations

With prior information like

$$B_l \preceq \Lambda \preceq B_u$$

with  $B_l, B_u \in \mathbf{S}_{++}^n$  we have a convex constraint which can be incorporated when we minimize  $h(\lambda, \Lambda)$ .

Also an upper bound  $\kappa_{\max}$  on the condition number of  $\Lambda$  can be incorporated by noting that it is equivalent to the existence of  $u > 0$  such that  $ul \preceq \Lambda \preceq \kappa_{\max} ul$ .

Prior information can be included:

$$m = \alpha m_0 + (1 - \alpha) \frac{1}{N} \sum_{i=1}^N x_i \quad (56)$$

$$M = \beta M_0 + (1 - \beta) \frac{1}{N} \sum_{i=1}^N x_i x_i^T \quad (57)$$

with  $\alpha, \beta \in [0, 1]$ .



# Relative Entropy

Difference between distributions measured by *relative entropy* or equivalently *Kullback-Leibler divergence*. Let

$$\mathcal{D}_n = \left\{ p \in \mathbf{R}^{\mathbf{R}^n} : \int_{\mathbf{R}^n} p(x) dx = 1, p(x) \geq 0, \forall x \in \mathbf{R}^n \right\}$$

Define relative entropy  $D : \mathcal{D}_n \times \mathcal{D}_n \rightarrow \mathbf{R}$  by

$$D(p, q) = - \int_{\mathbf{R}^n} p(x) \ln \frac{q(x)}{p(x)} dx$$

which is a convex function of  $p$ , since  $q(x) \geq 0, \forall x$ .

## Gibb's Inequality

What  $p$  minimizes  $D(p, q)$ ?

Lagrangian  $L : \mathcal{D}_n \times \mathbf{R} \rightarrow \mathbf{R}$  with

$$L(p, \mu) = D(p, q) + \mu \left( 1 - \int_{\mathbf{R}^n} p(x) dx \right)$$

First variation of the Lagrangian with respect to  $p$ :

$$\delta L = - \int_{\mathbf{R}^n} \delta p \left( \ln \frac{q}{p} - 1 + \mu \right) dx$$

$\implies$

$$p(x) = e^{1-\mu} q(x)$$

$\cdot \int_{\mathbf{R}^n} p(x) dx = 1 \implies \mu = 1$  and hence

$$D(p, q) \geq 0$$

with equality if and only if  $p(x) = q(x)$  for almost all  $x$ .

However, not a metric, since in general  $D(p, q) \neq D(q, p)$ .

# Cross Entropy

Cross entropy defined as  $C : \mathcal{D}_n \times \mathcal{D}_n \rightarrow \mathbf{R}$ , where

$$C(p, q) = - \int_{\mathbf{R}^n} p(x) \ln q(x) dx$$

With  $H(p)$  entropy of  $p$

$$C(p, q) = D(p, q) + H(p)$$

Hence

$$C(p, q) \geq H(p)$$

with equality if and only if  $p(x) = q(x)$  for almost all  $x$ .

Also

$$C(p, q) = -E_p(\ln q)$$

where  $E_p : \mathcal{D}_n \rightarrow \mathbf{R}$  is defined by

$$E_p(f) = \int_{\mathbf{R}^n} f(x) p(x) dx$$

# Cross Entropy and ML estimation

If we consider  $q$  to be parameterized with a parameter  $\theta \in \mathbf{R}^m$ , i.e.  $q : \mathbf{R}^n \times \mathbf{R}^m \rightarrow \mathbf{R}_+$ , then the ML problem is equivalent to

$$\text{minimize}_{\theta} - \sum_{k=1}^N \ln q(x_k; \theta)$$

where  $x_k$ ,  $k \in \mathbf{N}_N$  are the observed data of  $x$ . If we assume that the distribution for  $x$  is given by  $p$ , then the objective function above is proportional to a sample estimate of the cross entropy.

# Latent Variables or Missing Data

**Difficult** ML problem with likelihood function  $\ell : \mathbf{R}^N \times \mathbf{R}^p \rightarrow \mathbf{R}_+$  based on observations  $y \in \mathbf{R}^N$  parameterized with  $\theta \in \mathbf{R}^p$ :

$$\text{maximize}_{\theta} \ln \ell(y; \theta)$$

With additional observations  $z \in \mathbf{R}^M$  the ML problem with  $(y, z)$  as observations **easy**.

# Infinite Dimensional Optimization Problem

Let  $r : \mathbf{R}^N \times \mathbf{R}^M \times \mathbf{R}^p \rightarrow \mathbf{R}_+$  be the joint distribution for  $(y, z)$  with parameter  $\theta$ .

Define conditional distribution  $s : \mathbf{R}^N \times \mathbf{R}^M \times \mathbf{R}^p \rightarrow \mathbf{R}_+$ :

$$s(y, z; \theta) = \frac{r(y, z; \theta)}{\ell(y; \theta)}$$

Consider arbitrary distribution  $q \in \mathcal{D}_M$ .

Define optimization problem

$$\text{maximize}_{\theta, q} \ln \ell(y; \theta) - D(q, s)$$

where  $D$  is the relative entropy. (This trivially has same solution as ML problem, and  $q = s$  is optimal)

# Block-Coordinate Ascent

It holds

$$\ln \ell(y; \theta) - D(q, s) = \ln \ell(y; \theta) + H(q) - C(q, s) = H(q) - C(q, r)$$

where  $H$  is entropy and  $C$  is cross-entropy.

## Iterate:

1. Fix  $\theta$  and optimize with respect to  $q$ .
2. Fix  $q$  and optimize with respect to  $\theta$ .

First step has trivial solution  $q = s$ . ( $s$  depends on old value of  $\theta$ )

Second step equivalent to

$$\text{maximize}_{\theta} Q(\theta)$$

where  $Q(\theta) = -C(s, r) = E_s(\ln r)$ . ( $r$  depends on new value of  $\theta$ )

# The Expectation Maximization (EM) Algorithm

## Iterate:

1. Form  $Q(\theta) = E_s(\ln r)$  (E-step)
2. Solve  $\text{maximize}_{\theta} Q(\theta)$  (M-step)

Sometimes E-step is approximated by an empirical cross-entropy, i.e.

$$E_s(\ln r) \approx \frac{1}{M} \sum_{i=1}^M \ln r_i$$

where  $r_i = r(y, z_i; \theta)$  from drawing samples  $z_i$  from the conditional distribution  $s$ . This is called *Monte Carlo EM*.



# Gibbs Sampling

How to draw samples from a distribution  $p : \mathcal{D}_1 \times \cdots \times \mathcal{D}_n \rightarrow \mathbf{R}_+$ ?

Define the conditional distribution functions  $p_{i|\setminus i} : \mathcal{D}_i \rightarrow \mathbf{R}_+$ ,  $i \in \mathbf{N}_n$  as the distribution for  $x_i \in \mathcal{D}_i$  given

$$x_{\setminus i} = (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n).$$

Let  $x^{(0)} = (x_1^{(0)}, \dots, x_n^{(0)}) \in \mathcal{D}_1 \times \cdots \times \mathcal{D}_n$  be given. Then for  $k \in \mathbf{N}$  we compute  $x_i^{(k+1)}$  by drawing a sample from  $p_{i|\setminus i}(x_1^{(k+1)}, \dots, x_{i-1}^{(k+1)}, x_i, x_{i+1}^{(k)}, \dots, x_n^{(k)})$ .

Instead of cycling through the components make a random permutation of them for each  $k$ .

$x^{(k)}$  converges to a sample of  $p$  as  $k$  goes to infinity.

# Boltzmann Machine

Ising distribution:

$$p(x) = \exp(-E(x) - A(\lambda, \Lambda))$$

where  $E(x) = -\lambda^T x - \text{tr} \Lambda x x^T$  and  $A(\lambda, \Lambda) = \ln \sum_x \exp(-E(x))$  for  $x \in \mathcal{D}_x$ .

Divide  $x$  into *visible* and *hidden* or latent variables, i.e.  $x = (v, h)$ .

Consider  $N$  observations  $v_i$ ,  $i \in \mathbf{N}_N$  of  $v$ .

Corresponding hidden variables are  $h_i$ ,  $i \in \mathbf{N}_N$ .

Define  $x_i = (v_i, h_i)$ , and with abuse of notation  $v = (v_1, \dots, v_N)$ ,  $h = (h_1, \dots, h_N)$  and  $x = (v, h)$ .

# Likelihood Function

Define  $r : \mathcal{D}_x^N \times \mathbf{R}^m \times \mathbf{S}^m \rightarrow \mathbf{R}$  via

$$r(x, \lambda, \Lambda) = \prod_{i=1}^N p(x_i) = e^{-\bar{E}(x, \lambda, \Lambda) - N\Lambda(\lambda, \Lambda)}$$

where  $\bar{E} : \mathcal{D}_x^N \times \mathbf{R}^m \times \mathbf{S}^m \rightarrow \mathbf{R}$  is given by  
 $\bar{E}(x, \lambda, \Lambda) = \sum_{i=1}^N E(x_i)$ .

Conditional distribution for  $h$  given  $v$  is  
 $s : \mathcal{D}_v^N \times \mathcal{D}_h^N \times \mathbf{R}^m \times \mathbf{S}^m \rightarrow \mathbf{R}$  given by

$$s(v, h, \lambda, \Lambda) = \frac{r(v, h, \lambda, \Lambda)}{\sum_h r(v, h, \lambda, \Lambda)} = \frac{e^{-\bar{E}(v, h, \lambda, \Lambda)}}{\sum_h e^{-\bar{E}(v, h, \lambda, \Lambda)}}$$

where  $\mathcal{D}_v$  and  $\mathcal{D}_h$  are defined such that  $\mathcal{D}_v \times \mathcal{D}_v = \mathcal{D}_x$ .

## Q-function in EM Algorithm

Define  $Q : \mathbf{R}^m \times \mathbf{S}^m \rightarrow \mathbf{R}$  via

$$\begin{aligned} Q(\lambda, \Lambda) &= E_s \ln r = \sum_h s(v, h, \lambda^-, \Lambda^-) (-\bar{E}(v, h, \lambda, \Lambda) - NA(\lambda, \Lambda)) \\ &= - \sum_h s(v, h, \lambda^-, \Lambda^-) \bar{E}(v, h, \lambda, \Lambda) - NA(\lambda, \Lambda) \end{aligned}$$

where  $(\lambda^-, \Lambda^-)$  values of the parameters from the previous iteration in the EM algorithm.

Define the conditional distribution for  $h_i$  given  $v_i$  as

$s_i : \mathcal{D}_v \times \mathcal{D}_h \times \mathbf{R}^m \times \mathbf{S}^m \rightarrow \mathbf{R}$  via

$$s_i(v_i, h_i, \lambda, \Lambda) = \frac{p(x_i)}{\sum_{h_i} p(x_i)} = \frac{e^{-E(v_i, h_i, \lambda, \Lambda)}}{\sum_{h_i} e^{-E(v_i, h_i, \lambda, \Lambda)}}$$

# Gradients

$$\frac{\partial Q}{\partial \lambda} = - \sum_h s(h, v, \lambda^-, \Lambda^-) \frac{\partial \bar{E}}{\partial \lambda} - N \frac{\partial A}{\partial \lambda}$$

$$= \sum_h s(h, v, \lambda^-, \Lambda^-) \sum_{i=1}^N x_i - N \sum_{\xi} \xi p(\xi)$$

$$= \sum_{i=1}^N \sum_{h_i} s_i(h_i, v_i, \lambda^-, \Lambda^-) x_i - N \sum_{\xi} \xi p(\xi)$$

$$\frac{\partial Q}{\partial \Lambda} = \sum_{i=1}^N \sum_{h_i} s_i(h_i, v_i, \lambda^-, \Lambda^-) x_i x_i^T - N \sum_{\xi} \xi \xi^T p(\xi)$$

# Principal Component Analysis

Given data  $x_i \in \mathbf{R}^n$ ,  $i \in \mathbf{N}_N$  should be *approximated* with  $W^T c_i$ , where  $W \in \mathbf{R}^{m \times n}$ ,  $m < n$ , and where  $c_i \in \mathbf{R}^m$ ,  $i \in \mathbf{N}_N$ . Require  $WW^T = I$ .

Let  $J : \mathbf{R}^{m \times n} \times \mathbf{R}^n \times \cdots \times \mathbf{R}^n \rightarrow \mathbf{R}_+$  be defined by

$$J(W, c_1, \dots, c_N) = \frac{1}{2} \sum_{i=1}^N \|x_i - W^T c_i\|_2^2$$

Optimization problem:

$$\text{minimize}_{W, c_1, \dots, c_N} J(W, c_1, \dots, c_N) \quad (58)$$

$$\text{subject to } WW^T = I \quad (59)$$

Convex problem for  $c_i$  when  $W$  fixed.

## Optimization of $c_i$

$$\frac{\partial J}{\partial c_i} = WW^T c_i - Wx_i = 0, \quad i \in \mathbf{N}_N$$

$$WW^T = I \Rightarrow c_i = Wx_i \Rightarrow$$

$$\begin{aligned} J(L, Wx_1, \dots, Wx_N) &= \frac{1}{2} \sum_{i=1}^N x_i^T (I - W^T W)^2 x_i \\ &= \frac{1}{2} \sum_{i=1}^N x_i^T (I - W^T W) x_i \end{aligned}$$

Equivalent optimization problem:

$$\text{maximize}_W \frac{1}{2} \text{tr} WX^T XW^T \quad (60)$$

$$\text{subject to } WW^T = I \quad (61)$$

where

$$X^T = [x_1 \quad \dots \quad x_N]$$

## Equivalent Problem

Let  $X = UDV^T$  be an SVD and define  $Y = V^T W^T$  and  $\tilde{J}: \mathbf{R}^{n \times m} \rightarrow \mathbf{R}_+$  where

$$\tilde{J}(Y) = \frac{1}{2} \text{tr } Y^T D^2 Y$$

and equivalent problem

$$\text{maximize}_Y \tilde{J}(Y) \tag{62}$$

$$\text{subject to } Y^T Y = I \tag{63}$$

Lagrangian  $L: \mathbf{R}^{n \times m} \times \mathbf{S}^m \rightarrow \mathbf{R}$  where

$$L(Y, \Lambda) = \tilde{J}(Y) + \text{tr } \Lambda (I - Y^T Y)$$

Necessary condition for optimal  $Y$ :  $\exists \Lambda$ :

$$\frac{\partial L}{\partial Y} = D^2 Y - Y \Lambda = 0$$



# Optimality Conditions

Let  $Z$  be such that  $\begin{bmatrix} Y & Z \end{bmatrix}^T \begin{bmatrix} Y & Z \end{bmatrix} = I$ .

Then equivalent optimality condition:

$$Y^T D^2 Y - \Lambda = 0 \quad (64)$$

$$Z^T D^2 Y = 0 \quad (65)$$

First equation always has a solution  $\Lambda$  for any  $Y$ .

Hence the necessary conditions are  $\exists Z$ :

$$\begin{bmatrix} Y & Z \end{bmatrix}^T \begin{bmatrix} Y & Z \end{bmatrix} = I \quad (66)$$

$$Z^T D^2 Y = 0 \quad (67)$$

One solution is  $\begin{bmatrix} Y & Z \end{bmatrix} = I$ . Also  $\begin{bmatrix} Y & Z \end{bmatrix} = X_1 \oplus X_2$  for any orthogonal  $X_1 \in \mathbf{R}^{m \times m}$  and  $X_2 \in \mathbf{R}^{(n-m) \times m}$ .

# Proof of Optimality

Objective function:

$$\frac{1}{2} \sum_{i=1}^N \|Dy_i\|_2^2$$

where  $Y = [y_1 \ \cdots \ y_m]$ , and

$$Dy_1 = \begin{bmatrix} d_1 y_{11} \\ d_2 y_{21} \\ \vdots \\ d_n y_{n1} \end{bmatrix}$$

Optimize with respect to  $y_1$  first. Since  $d_i \geq d_j$  when  $i < j$  and  $y_i^T y_i = 1$  it is optimal to take  $y_1 = e_1$ .

Remaining  $y_i$  for  $2 \leq i \leq m$  has to have its first components equal to zero in order to be orthogonal to  $y_1$ .

Because of this  $y_2 = e_2$  and so on  $\Rightarrow Y = \begin{bmatrix} I \\ 0 \end{bmatrix}$  optimal.

## Non-Uniqueness

With  $Y = \begin{bmatrix} X_1 \\ 0 \end{bmatrix}$  where  $X_1$  orthogonal it holds

$$\begin{aligned}\tilde{J}\left(\begin{bmatrix} X_1 \\ 0 \end{bmatrix}\right) &= \frac{1}{2} \mathbf{tr} \left( \begin{bmatrix} X_1 \\ 0 \end{bmatrix}^T D^2 \begin{bmatrix} X_1 \\ 0 \end{bmatrix} \right) = \frac{1}{2} \mathbf{tr} \left( X_1^T D_1^2 X_1 \right) \\ &= \frac{1}{2} \mathbf{tr} X_1 X_1^T D_1^2 = \frac{1}{2} \mathbf{tr} D_1^2\end{aligned}$$

where  $D = D_1 \oplus D_2$ .

Hence the principal component analysis picks out the components of  $x_i$  corresponding to the  $m$  largest singular values of  $X^T$ .

# Low Rank Approximation

We approximate  $x_i$  with  $W^T c_i$ , where  $c_i = Wx_i$ . Because of this  $X$  is approximated with  $XW^T W$ .

Since  $W$  is an orthogonal matrix,  $W^T W$  is a projection matrix, and hence **Rank**  $XW^T W = m$ .

We have  $XW^T W = UDV^T V_1 V_1^T = UD \begin{bmatrix} I \\ 0 \end{bmatrix} V_1^T = U_1 D_1 V_1^T$ ,  
where  $U = [U_1 \ U_2]$  and  $V = [V_1 \ V_2]$ .

## Mutual Information

For joint distribution  $r : \mathbf{R}^m \times \mathbf{R}^n \rightarrow \mathbf{R}_+$  with marginal distributions  $p : \mathbf{R}^m \rightarrow \mathbf{R}_+$  and  $q : \mathbf{R}^n \rightarrow \mathbf{R}_+$  mutual information  $I : \mathcal{D}_{m+n} \rightarrow \mathbf{R}_+$  is

$$I(r) = \int_{\mathbf{R}^m \times \mathbf{R}^n} r(x, y) \ln \frac{r(x, y)}{p(x)q(y)} dx dy$$

Consider two zero mean  $n$ -dimensional independent random variables  $x$  and  $e$  with normal distributions with covariances  $\Sigma \in \mathbf{S}_+^n$  and  $I$ . Let  $z = Wx$  and  $y = z + e$ , where  $W \in \mathbf{R}^{m \times n}$ .

$(y, z)$  is zero mean normal distribution  $r$  with covariance

$$\begin{bmatrix} W\Sigma W^T + I & W\Sigma W^T \\ W\Sigma W^T & W\Sigma W^T \end{bmatrix}$$

We let  $p$  and  $q$  be the marginal distributions for  $y$  and  $z$  and define  $J : \mathbf{R}^{m \times n} \rightarrow \mathbf{R}_+$  as

$$J(W) = I(r) = \frac{1}{2} \ln \det \left( I + W\Sigma W^T \right)$$

# Optimization Problem

$$\text{maximize}_W J(W) \tag{68}$$

$$\text{subject to } WW^T = I \tag{69}$$

Constraint will make objective bounded.

Let  $V : V^T V = I$  and  $\Sigma = VD^2V^T$  with  $D$  diagonal. Then

$$J(W) = \frac{1}{2} \ln \det \left( I + Y^T D^2 Y \right)$$

where  $Y = V^T W^T$ . Assume  $d_i \geq d_j$  for all  $i < j$ .

# Equivalent Optimization Problem

$$\text{maximize}_Y \tilde{J}(Y) \quad (70)$$

$$\text{subject to } Y^T Y = I \quad (71)$$

where  $\tilde{J} : \mathbf{R}^{n \times m} \rightarrow \mathbf{R}_+$  with  $\tilde{J}(Y) = J(W)$ .

Lagrangian  $L : \mathbf{R}^{n \times m} \times \mathbf{S}^m \rightarrow \mathbf{R}$  where

$$L(Y, \Lambda) = \frac{1}{2} \ln \det \left( I + Y^T D^2 Y \right) + \frac{1}{2} \text{tr} \Lambda \left( I - Y^T Y \right)$$

The existence of  $\Lambda$  :

$$\frac{\partial L}{\partial Y} = D^2 Y \left( I + Y^T D^2 Y \right)^{-1} - Y \Lambda = 0$$

is necessary condition for optimality of  $Y$ .

## Equivalent Optimality Conditions

Let  $Z$  be s.t.  $\begin{bmatrix} Y & Z \end{bmatrix}$  is square and orthogonal.

Equivalent optimality conditions:  $\exists Z$  and  $\Lambda$ :

$$Y^T D^2 Y \left( I + Y^T D^2 Y \right)^{-1} - \Lambda = 0 \quad (72)$$

$$Z^T D^2 Y = 0 \quad (73)$$

with  $\begin{bmatrix} Y & Z \end{bmatrix}$  orthogonal. From  $A(I + A)^{-1} = I - (I + A)^{-1}$

$$I - \left( I + Y^T D^2 Y \right)^{-1} - \Lambda = 0 \quad (74)$$

$$Z^T D^2 Y = 0 \quad (75)$$

First equation has a solution in terms of  $\Lambda$  for any  $Y$ .

Optimality conditions:  $\exists Z \in \mathbf{R}^{n \times (n-m)}$  such that

$$Z^T D^2 Y = 0 \quad (76)$$

$$\begin{bmatrix} Y & Z \end{bmatrix}^T \begin{bmatrix} Y & Z \end{bmatrix} = I \quad (77)$$



## Relation to PCA

Same optimality conditions as for PCA if we identify  $\Sigma$  with  $X^T X$ .

However, the objective functions are not the same.

As in the previous section,  $\begin{bmatrix} Y & Z \end{bmatrix} = X_1 \oplus X_2$  with  $X_1 \in \mathbf{R}^{m \times m}$  and  $X_2 \in \mathbf{R}^{(n-m) \times (n-m)}$  orthogonal is solution to optimality conditions.

It holds

$$\tilde{J}(Y) = \frac{1}{2} \sum_{k \in \mathbf{N}_m} \ln(1 + d_k^2)$$

It can be shown using Givens rotations that no other orthogonal  $\begin{bmatrix} Y & Z \end{bmatrix}$  can be optimal.

## Non-orthogonal $y_i$

$$\text{maximize}_Y \tilde{J}(Y) \quad (78)$$

$$\text{subject to } \|y_i\|_2^2 = 1, \quad i \in \mathbf{N}_m \quad (79)$$

Lagrangian  $M : \mathbf{R}^{n \times m} \times \mathbf{R}^m \rightarrow \mathbf{R}$  where

$$M(Y, \lambda) = \frac{1}{2} \ln \det \left( I + Y^T D^2 Y \right) + \frac{1}{2} \sum_{i=1}^m \lambda_i (1 - \|y_i\|_2^2)$$

We see that

$$\frac{\partial M}{\partial Y} = D^2 Y \left( I + Y^T D^2 Y \right)^{-1} - Y \mathbf{diag}(\lambda)$$

and necessary conditions for optimality:  $\exists Z$  and  $\lambda \in \mathbf{R}^m$ :

$$Y^T D^2 Y \left( I + Y^T D^2 Y \right)^{-1} - Y^T Y \mathbf{diag}(\lambda) = 0 \quad (80)$$

$$Z^T D^2 Y = 0 \quad (81)$$

where  $Z$  is such that  $\begin{bmatrix} Y & Z \end{bmatrix}$  not necessarily square has full column rank.

## Equivalent optimality conditions

$$I - \left(I + Y^T D^2 Y\right)^{-1} - Y^T Y \mathbf{diag}(\lambda) = 0 \quad (82)$$

$$Z^T D^2 Y = 0 \quad (83)$$

$\begin{bmatrix} Y & Z \end{bmatrix} = I$  is a solution. However, any orthogonal  $\begin{bmatrix} Y & Z \end{bmatrix}$  will not satisfy them, since for orthogonal  $Y$  it must hold that  $I + Y^T D^2 Y$  is diagonal for there to exist  $\lambda$  that satisfies the equation.

There are however non-orthogonal  $Y$  that satisfy them. Consider  $Y = \begin{bmatrix} \mathbf{1} & 0 \end{bmatrix}^T$ . Then  $\lambda = d_1^2 / (1 + m d_1^2) \mathbf{1}$  satisfies the necessary optimality conditions.

It actually holds that we may take  $y_i$  equal to any basis vector for  $\mathbf{R}^n$ , and they may be linearly dependent.

# What Stationary Points are Optimal?

Example where  $m = 2$ ,  $d_1 > d_2$ .

For  $Y = [\mathbf{1} \ 0]^T$  we have

$\ln \det(I + Y^T D^2 Y) = \ln(1 + 2d_1^2) \approx 2d_1^2$  for small values of  $d_1^2$ .

For  $Y = \begin{bmatrix} I \\ 0 \end{bmatrix}$  we have

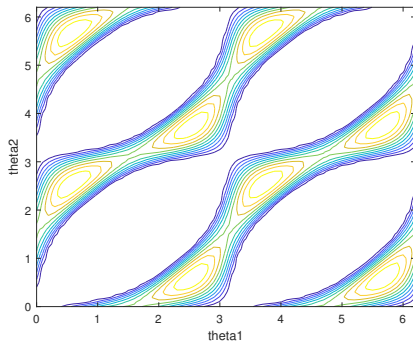
$\ln \det(I + Y^T D^2 Y) = \ln(1 + d_1^2 + d_2^2 + d_1^2 d_2^2) \approx d_1^2 + d_2^2 + d_1^2 d_2^2$   
for small values of  $d_1^2$  and  $d_2^2$ .

Exist  $d_1 > d_2$  for which second approximation is smaller than the first and for small SNR it is better to only consider the signal with the largest variance.

In general there are cases in-between, where one should pick out more than one of the largest, but not all  $m$ .

## Non-Aligned and Non-Orthogonal Case

Example with  $m = n = 2$  and let  $y_1 = \begin{bmatrix} \cos \theta_1 \\ \sin \theta_1 \end{bmatrix}$  and  $y_2 = \begin{bmatrix} \cos \theta_2 \\ \sin \theta_2 \end{bmatrix}$ .



Several optima: e.g.

$$Y = \begin{bmatrix} 0.83 & 0.83 \\ -0.56 & 0.56 \end{bmatrix}$$

which corresponds to  $\theta_1 = 5.7$  and  $\theta_2 = 0.6$ . Other optima have same angle between  $y_1$  and  $y_2$ .

## Saddle Points and Large SNR

$$Y = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}; \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}; \quad \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

correspond to saddle points.

If the signal to noise ratio is large, then  $Y = \begin{bmatrix} X_1 \\ 0 \end{bmatrix}$  with  $X_1$

orthogonal is optimal also without orthogonality constraints, since first equation of optimality conditions in the limit when  $D$  is much larger than  $I$  is given by

$$I - Y^T Y \mathbf{diag}(\lambda) = 0$$

## Relationship to PCA

Assume observations  $x_i$ ,  $i \in \mathbf{N}_N$  of  $x$  that we want to approximate with  $W^T z_i$ , where  $z_i = Wx_i$ . We estimate the covariance matrix of  $x$  with  $\Sigma = \frac{1}{N} X^T X$ , where

$$X = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_N^T \end{bmatrix}$$

We then let  $X = UDV^T$  be a singular value decomposition of  $X$ . Then

$$J(W) = \frac{1}{2} \ln \det \left( I + Y^T \frac{D^2}{N} Y \right)$$

where  $Y = V^T W^T$ .

The approximation of  $X$  will be  $XW^T W$ , where  $W^T W$  is not necessarily a projection matrix, but rank of  $XW^T W$  will still be  $m$ .

# Why Mutual Information is Superior to PCA

Notice that we cannot relax the condition on orthogonality in the principal component analysis without obtaining  $Y = \begin{bmatrix} \mathbf{1}^T \\ 0 \end{bmatrix}$  as the optimal solution.

The signal to noise ratio will not help in making the correct choice from an information point of view. This is why optimizing mutual information seems to be more appropriate.



# Cluster Analysis

Partition observations  $x_i \in \mathbf{R}^n$ ,  $i \in \mathbf{N}_N$  into groups called *clusters* so that the observations in each cluster are close to one another.

We may measure distance  $d : \mathbf{R}^n \times \mathbf{R}^n \rightarrow \mathbf{R}_+$  as e.g.  
 $d(x_i, x_j) = \|x_i - x_j\|^2$ .

Assume  $K < N$  clusters and define *encoder*  $C : \mathbf{N}_N \rightarrow \mathbf{N}_K$  that assigns each observation to each cluster.

The sum of all distances within each cluster should be minimized:

$$\text{minimize}_C f(C)$$

where  $f : \mathbf{N}_K \rightarrow \mathbf{R}_+$  is defined by

$$f(C) = \frac{1}{2} \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \sum_{j \in \mathcal{C}_k} d(x_i, x_j)$$

where  $\mathcal{C}_k = \{i \in \mathbf{N}_N : C(i) = k\}$ .

## Equivalent Formulation

Define the mean vectors

$$m_k = \frac{1}{N_k} \sum_{i \in \mathcal{C}_k} x_i, \quad k \in \mathbf{N}_K$$

where  $N_k = |\mathcal{C}_k|$ . Then if  $d$  is the squared Euclidian norm:

$$f(C) = \frac{1}{2} \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \sum_{j \in \mathcal{C}_k} d(x_i, x_j) = \sum_{k=1}^K N_k \sum_{i \in \mathcal{C}_k} d(x_i, m_k)$$

We have for any  $y_k \in \mathbf{R}^n$  that  $\sum_{i \in \mathcal{C}_k} d(x_i, m_k) \leq \sum_{i \in \mathcal{C}_k} d(x_i, y_k)$  with equality for  $y_k = m_k$  and hence an equivalent optimization:

$$\text{minimize}_{C, y_1, \dots, y_K} F(C, y_1, \dots, y_K)$$

where  $F : \mathbf{N}_N \times \mathbf{N}_K \times \mathbf{R}^n \times \dots \times \mathbf{R}^n \rightarrow \mathbf{R}_+$  is defined as

$$F(C, y_1, \dots, y_K) = \sum_{k=1}^K N_k \sum_{i \in \mathcal{C}_k} d(x_i, y_k)$$

Sets  $\mathcal{C}_k$  are functions of encoder  $C$ .

# K-Means Algorithm

Block coordinate descent:

1. For fixed  $C$  solve  $\text{minimize}_{y_1, \dots, y_K} F(C, y_1, \dots, y_K)$
2. For fixed  $(y_1, \dots, y_K)$  solve  $\text{minimize}_C F(C, y_1, \dots, y_K)$

First problem is a LS problem.

Second problem has an explicit solution given by assigning observation  $x_i$  to cluster  $k$  if  $d(x_i, y_k) \leq d(x_i, y_j)$  for all  $j \neq k$ .

The algorithm can be trapped in local minima.

## Regression

Fit polynomial function  $y : \mathbf{R} \rightarrow \mathbf{R}$  where

$$y(x) = a_1 + a_2x + \cdots a_mx^{m-1}$$

to pairs  $(x_k, y_k)$  for  $i \in \mathbf{N}_N$ .

Define  $a = (a_1, \dots, a_m) \in \mathbf{R}^m$  and  $\beta : \mathbf{R} \rightarrow \mathbf{R}^m$  by  $\beta(x) = (1, x, \dots, x^{m-1})$ . Then

$$y(x) = a^T \beta(x)$$

called a *linear regression* model.

Fit by solving

$$\text{minimize}_a \frac{1}{2} \sum_{k=1}^N \left( a^T \beta(x_k) - y_k \right)^2 \quad (84)$$

This is a linear Least Squares (LS) problem

# Generalization

General functions  $\varphi_j : \mathbf{R} \rightarrow \mathbf{R}$ ,  $j \in \mathbf{N}_m$  and let

$$y(x) = \sum_{j=1}^m a_j \varphi_j(x)$$

Often  $\varphi_1(x) = 1$ ,  $\forall x$ . With  $\beta(x) = (\varphi_1(x), \dots, \varphi_m(x))$ .

$$y(x) = a^T \beta(x)$$

# Vector-Valued Regression Model

Let  $y : \mathbf{R}^n \rightarrow \mathbf{R}^p$  be given by

$$y(x) = A\beta(x)$$

where  $A \in \mathbf{R}^{p \times m}$ ,  $\beta : \mathbf{R}^n \rightarrow \mathbf{R}^m$  with  $\beta(x) = (\varphi_1(x), \dots, \varphi_m(x))$  and  $\varphi_j : \mathbf{R}^n \rightarrow \mathbf{R}$ ,  $j \in \mathbf{N}_m$ . Important special case:  $\beta(x) = x$ .

LS criterion is the sum of the LS criteria for each row in regression model, or

$$\text{minimize}_A \frac{1}{2} \sum_{k=1}^N \text{tr} (y_k - A\beta(x_k)) (y_k - A\beta(x_k))^T$$

The solution to this problem is closely related to the affine predictor.

# Statistical Interpretation

Let  $e_k$  be independent with distribution  $p : \mathbf{R} \rightarrow \mathbf{R}_+$  where

$$p(e) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}e^2}$$

Let

$$y(x_k) = a^T \beta(x_k) + e_k$$

with  $a \in \mathbf{R}^m$  and  $\beta : \mathbf{R}^n \rightarrow \mathbf{R}^m$  model the observations  $y_k$ ,  $k \in \mathbf{N}_N$ .

Likelihood function  $\ell : \mathbf{R}^N \times \mathbf{R}^m \rightarrow \mathbf{R}_+$

$$\ell(y, \dots y_N; a) = \prod_{k=1}^N p(y_k - a^T \beta(x_k))$$

Then

$$-\ln \ell(y_1, \dots y_N; a) = \sum_{k=1}^N \frac{1}{2\sigma^2} (y_k - a^T \beta(x_k))^2 + N \ln(\sqrt{2\pi}\sigma)$$

Minimizing  $a$  same as LS solution.

# Maximum A Posteriori (MAP) Estimation

Assume that also  $a$  is a random variable with some distribution  $q : \mathbf{R}^m \rightarrow \mathbf{R}_+$  independent of  $e_k$ .

Joint distribution  $r : \mathbf{R}^N \times \mathbf{R}^m \rightarrow \mathbf{R}_+$  of the observations  $y = (y_1, \dots, y_N)$  and parameter  $a$ :

$$r(y; a) = q(a) \prod_{k=1}^N p(y_k - a^T \beta(x_k))$$

which is proportional to distribution of  $a$  conditioned on the observations  $y$ .

$$\text{minimize}_a \sum_{k=1}^N \frac{1}{2\sigma^2} (y_k - a^T \beta(x_k))^2 + N \ln(\sqrt{2\pi}\sigma) - \ln q(a)$$



# Regularization

Different distributions  $q$  result in different regularizations.

Normal distribution with zero mean and covariance  $\Sigma$ :

$$-\ln q(a) \sim a^T \Sigma^{-1} a$$

Called *Ridge regression* or *Tikhonov regularization*.

Components  $a_i$  of  $a$  have double-sided exponential distribution:

$$q(a) = \prod_{i=1}^m \frac{1}{2\lambda} e^{-|a_i|/\lambda} \Rightarrow -\ln q(a) \sim \sum_{i=1}^m |a_i|$$

Called *Lasso regularization*.

Any regularization function  $h_i(a_i)$  can be translated into a distribution for  $a_i$  by the formula  $q_i(a_i) = e^{-h_i(a_i)}/\mu$ , where  $\mu > 0$  is a normalization constant.

# Constrained Regression

Consider

$$q_i(a_i) = \begin{cases} \frac{1}{2\lambda}, & a_i \in [-\lambda, \lambda] \\ 0 & a_i \notin [-\lambda, \lambda] \end{cases}$$
$$\implies$$

$$-\ln q_i(a_i) = \begin{cases} \ln(2\lambda), & a_i \in [-\lambda, \lambda] \\ +\infty & a_i \notin [-\lambda, \lambda] \end{cases}$$

Hence the terms  $-\ln q_i(a_i)$  in the objective function of the MAP problem can be replaced with the constraint  $a_i \in [-\lambda, \lambda]$  for  $i \in \mathbf{N}_m$  or equivalently  $\|a\|_\infty \leq \lambda$ .

# Hilbert Spaces

For  $a = (a_1, a_2, \dots)$ , where  $a_i \in \mathbf{R}$ ,  $i \in \mathbf{N}$  we say that  $a \in \ell_2$  if  $\sum_{i=1}^{\infty} a_i^2 < \infty$ .

Inner product:  $\langle \cdot, \cdot \rangle_{\ell_2} : \ell_2 \times \ell_2 \rightarrow \mathbf{R}$  by  $\langle a, b \rangle_{\ell_2} = \sum_{i=1}^{\infty} a_i b_i$ .

Corresponding norm:  $\| \cdot \|_{\ell_2} \rightarrow \mathbf{R}_+$  is defined as  $\|a\|_{\ell_2}^2 = \langle a, a \rangle_{\ell_2}$ .

For  $f : \mathbf{R}^n \rightarrow \mathbf{R}$  we say that  $f \in L_2$  if

$$\int_{\mathbf{R}^n} f(x)^2 dx < \infty$$

Inner product:  $\langle \cdot, \cdot \rangle_{L_2} : L_2 \times L_2 \rightarrow \mathbf{R}$  defined as

$$\langle f, g \rangle_{L_2} = \int_{\mathbf{R}^n} f(x)g(x)dx$$

Corresponding norm:  $\| \cdot \|_{L_2} : L_2 \rightarrow \mathbf{R}_+$  is defined as  $\|f\|_{L_2}^2 = \langle f, f \rangle_{L_2}$ .

# Regression in Hilbert Spaces

Let  $\beta(x) = (\varphi_1(x), \varphi_2(x), \dots)$ , where  $\varphi_i : \mathbf{R}^n \rightarrow \mathbf{R}$ ,  $i \in \mathbf{N}$ .

Assume  $\varphi_i \in L_2$ ,  $i \in \mathbf{N}$ :

$$\begin{aligned}\langle \varphi_i, \varphi_j \rangle_{L_2} &= 0, & i &\neq j \\ \langle \varphi_i, \varphi_j \rangle_{L_2} &= 1, & i &= j\end{aligned}$$

Let  $f(x) = \sum_{i=1}^{\infty} a_i \varphi_i(x)$  be a regressor and consider

$$\text{minimize}_a \frac{1}{2} \sum_{k=1}^N (y_k - f(x_k))^2 + \frac{\nu}{2} \|f\|_{L_2}^2$$

# Equivalent Optimization Problem

A series  $f(x) = \sum_{i=1}^{\infty} a_i \varphi_i(x)$  is convergent and belongs to  $L_2$  for orthonormal  $\varphi_i$  if and only if  $a \in \ell_2$ . Then  $a_i = \langle f, \varphi_i \rangle_{L_2}$  and hence  $\|f\|_{L_2} = \|a\|_{\ell_2}$ .

We reformulate this problem as a constrained problem:

$$\begin{aligned} & \text{minimize}_{a,e} \quad \frac{1}{2} \sum_{k=1}^N e_k^2 + \frac{\nu}{2} \|a\|_{\ell_2}^2 \\ & \text{subject to } e_k = y_k - \sum_{i=1}^{\infty} a_i \varphi_i(x), \quad k \in \mathbf{N}_K \end{aligned}$$

where  $e = (e_1, \dots, e_N)$ .

# Vector Notation

To ease notation:

$$\begin{aligned} &\text{minimize}_{a,e} \frac{1}{2} e^T e + \frac{\nu}{2} a^T a \\ &\text{subject to } e = y - \Phi a \end{aligned}$$

where  $\Phi$  is infinite-dimensional matrix:

$$\Phi = \begin{bmatrix} \beta^T(x_1) \\ \vdots \\ \beta^T(x_N) \end{bmatrix}$$

# Lagrangian

Lagrangian  $L : \mathbf{R}^N \times \ell_2 \times \mathbf{R}^N \rightarrow \mathbf{R}$  defined by

$$L(e, a, \lambda) = \frac{1}{2}e^T e + \frac{\nu}{2}a^T a + \lambda^T (e - y + \Phi a)$$

Completing the squares:

$$\begin{aligned} L(e, a, \lambda) &= \frac{1}{2}(e + \lambda)^T (e + \lambda) + \frac{\nu}{2} \left( a + \frac{1}{\nu} \Phi^T \lambda \right)^T \left( a + \frac{1}{\nu} \Phi^T \lambda \right) \\ &\quad - \frac{1}{2} \lambda^T \lambda - \frac{1}{2\nu} \lambda^T \Phi \Phi^T \lambda - \lambda^T y \end{aligned}$$

Minimized for

$$e = -\lambda; \quad a = -\frac{1}{\nu} \Phi^T \lambda$$

## Kernel Trick

Lagrange dual function  $g : \mathbf{R}^N \rightarrow \mathbf{R}$  defined by

$$g(\lambda) = -\frac{1}{2}\lambda^T \lambda - \frac{1}{2\nu}\lambda^T \Phi \Phi^T \lambda - \lambda^T y$$

Kernel function  $K : \mathbf{R}^n \times \mathbf{R}^n \rightarrow \mathbf{R}$  defined by

$$K(x, \bar{x}) = \beta(x)^T \beta(\bar{x}) = \sum_{i=1}^{\infty} \varphi_i(x) \varphi_i(\bar{x})$$

Define  $\mathcal{K} \in \mathbf{S}^N$  with elements  $\mathcal{K}_{i,j} = K(x_i, x_j)$ . Then dual optimization problem

$$\text{minimize}_{\lambda} \frac{1}{2} \lambda^T \left( I + \frac{1}{\nu} \mathcal{K} \right) \lambda + \lambda^T y$$

with solution

$$\lambda = - \left( I + \frac{1}{\nu} \mathcal{K} \right)^{-1} y$$

We may also write

$$f(x) = a^T \beta(x) = -\frac{1}{\nu} \lambda^T \Phi \beta(x) = -\frac{1}{\nu} \sum_{k=1}^N \lambda_k K(x_k, x)$$



# Kernel Functions

*Positive semidefinite kernel:* for any  $N \in \mathbf{N}$ , and any  $c_i \in \mathbf{R}$ , and  $x_i \in \mathbf{R}^n$ ,  $i \in \mathbf{N}_N$  it should hold that

$$\sum_{i=1}^N \sum_{j=1}^N c_i c_j K(x_i, x_j) \geq 0$$

*Mercer's theorem:* If  $K$  is a positive definite kernel, then there are orthonormal  $\varphi_i$  that defines it. (could be finitely or infinitely many)

*dth degree polynomials:*  $K(x, \bar{x}) = (1 + x^T \bar{x})^d$

*Radial basis functions:* functions that only depend on  $\|x - \bar{x}\|_2$

*Gaussian radial basis kernel*

$$K(x, \bar{x}) = \exp \left( -\frac{1}{2\sigma^2} \|x - \bar{x}\|_2^2 \right)$$

where  $\sigma \in \mathbf{R}_{++}$ .

# Gaussian Linear Regression

Linear regression model:

$$y_k = a^T x_k + e_k, \quad k \in \mathbf{N}_N$$

where  $y_k \in \mathbf{R}$ ,  $x_k \in \mathbf{R}^n$ , where  $a$  zero mean random variable with a normal distribution and covariance  $\Sigma_a \in \mathbf{S}_{++}^n$ , and where  $e = (e_1, \dots, e_N)$  zero mean random variable with normal distribution and covariance  $\Sigma_e \in \mathbf{S}_{++}^N$ .

Let  $y = (y_1, \dots, y_N)$  and

$$X = \begin{bmatrix} x_1^T \\ \vdots \\ x_N^T \end{bmatrix}$$

Then

$$y = Xa + e$$

# Conditional Distribution

Given observations  $y$  and  $X$  the mean and the covariance of the posterior normal distribution are

$$\hat{a} = \Sigma_{a|y} X^T \Sigma_e^{-1} y = \Sigma_a X^T \left( \Sigma_e + X \Sigma_a X^T \right)^{-1} y$$
$$\Sigma_{a|y} = \left( \Sigma_a^{-1} + X^T \Sigma_e^{-1} X \right)^{-1}$$

Prediction with a new value  $x$ :

$$\hat{a}^T x = x^T \hat{a} = x^T \Sigma_a X^T \left( \Sigma_e + X \Sigma_a X^T \right)^{-1} y$$

# Kernel Function Interpretation

Notice that  $\Sigma_a$  only appears in terms of expressions of the form  $z^T \Sigma_a \bar{z}$  for  $z \in \mathbf{R}^n$  and  $\bar{z} \in \mathbf{R}^n$ .

Can instead write predictor in terms of kernel function  $K : \mathbf{R}^n \times \mathbf{R}^n \rightarrow \mathbf{R}_+$  defined by  $K(z, \bar{z}) = z^T \Sigma_a \bar{z}$ .

Let  $\mathcal{K} \in \mathbf{S}_+^N$ :  $\mathcal{K}_{i,j} = K(x_i, x_j)$ , and let  $\eta_k = K(x_k, x)$ ,  $k \in \mathbf{N}_N$ .  
Then

$$\hat{a}^T x = \sum_{k=1}^N \eta_k (\Sigma_e + \mathcal{K})^{-1} y$$

# Hilbert Space Regression Interpretation

If  $\Sigma_e = \nu I$  and if  $\varphi_i$  orthonormal such that resulting kernel function is  $K(z, \bar{z}) = z^T \Sigma_a \bar{z}$ , then above predictor is same as for Hilbert space regression.

$K(z, \bar{z})$  is covariance between  $a^T z$  and  $a^T \bar{z}$ .

# Gaussian Processes

Generalization of regressor:

$$y_k = f(x_k) + e_k$$

with  $f : \mathbf{R}^n \rightarrow \mathbf{R}$ , where we specify the covariance between  $f(z)$  and  $f(\bar{z})$  for any  $z \in \mathbf{R}^n$  and  $\bar{z} \in \mathbf{R}^n$  by specifying the kernel function.

Still assume that the joint distribution is Gaussian. This defines a zero mean real-valued *Gaussian random process* on  $\mathbf{R}^n$ .

# Kernel Functions

Any of the previous kernel functions may be used.

The Gaussian radial basis function is called the squared exponential covariance function.

Another common covariance function:

$$K(z, \bar{z}) = \exp \left( \frac{1}{\sigma} \|z - \bar{z}\|_2 \right)$$

which defines the *Ornstein–Uhlenbeck* process, where  $\sigma \in \mathbf{R}_{++}$ .

# Properties of Random Processes

When covariance function only depends on  $z - \bar{z}$ , the process is *stationary*.

When it only depends on  $\|z - \bar{z}\|_2$  is also *isotropic*.

Stationary and isotropic together is called *homogeneous*.

We specifically realize that infinite dimensional regression in Hilbert spaces can be equivalently expressed as MAP estimation for Gaussian processes.



# Classification

Pairs of data  $(y_k, x_k)$  for  $k \in \mathbf{N}_N$  with  $x_k \in \mathbf{R}^n$ , where  $y_k \in \mathbf{N}_K$  are called *qualitative, categorical, discrete* or *factors*.

We say that  $(y_k, x_k)$  belongs to class  $l$  if  $y_k = l$ .

Determine functions  $f_l : \mathbf{R}^n \rightarrow \mathbf{R}$ ,  $l \in \mathbf{N}_K$ :  $f_l(x_k) > 0$  if  $y_k = l$  and  $f_l(x_k) < 0$  if  $y_k \neq l$ .

The set  $\{x \mid f_l(x) = 0\}$  separates class  $l$  from the other classes.

# Classification using Linear Regression

Model each class  $l \in \mathbf{N}_K$  as a linear regression

$$z_l = a_l^T x + b_l$$

where  $a_l \in \mathbf{R}^n$  and  $b_l \in \mathbf{R}$ .

LS problem:

$$\text{minimize}_{a,b} \frac{1}{2} \sum_{k:y_k=l} \left( a_l^T x_k + b_l - 1 \right)^2 + \sum_{k:y_k \neq l} \left( a_l^T x_k + b_l \right)^2$$

where  $a = (a_1, \dots, a_K)$  and  $b = (b_1, \dots, b_K)$ .

*Discriminant functions:*  $\delta_l : \mathbf{R}^n \rightarrow \mathbf{R}$ :  $\delta_l(x) = a_l^T x + b_l$ .

Classify  $x$  to belong to class  $l$  if  $\delta_l(x) > \delta_k(x)$  for all  $k \neq l$ . Hence  $f_l(x) = \delta_l(x) - \max_{k \neq l} \delta_k(x)$ .

Prone to give bad results for  $K \geq 3$ . More general basis functions better.

# Logistic Regression

*Categorical distribution:*

$$p_l(z_1, z_2, \dots, z_{K-1}) = \begin{cases} \frac{e^{z_l}}{1 + \sum_{k=1}^{K-1} e^{z_k}}, & l \in \mathbf{N}_{K-1} \\ \frac{1}{1 + \sum_{k=1}^{K-1} e^{z_k}}, & l = K \end{cases}$$

where  $p_l : \mathbf{R}^{K-1} \rightarrow [0, 1]$  for  $l \in \mathbf{N}_K$ . Let  $z_l = a_l^T x + b_l$  and define likelihood function  $\ell : \mathbf{R}^{Nn} \times \mathbf{R}^{(K-1)(n+1)} \rightarrow [0, 1]$ :

$$\begin{aligned} \ell(x_1, \dots, x_N; a_1, \dots, a_{K-1}, b_1, \dots, b_{K-1}) \\ = \prod_{k: y_k = l} p_l(a_1^T x_k + b_1, \dots, a_{K-1}^T x_k + b_{K-1}) \end{aligned}$$

Log-likelihood function:

$$\sum_{k: y_k = l, l \in \mathbf{N}_{K-1}} a_l^T x_k + b_l - \sum_{k=1}^N \ln \left( 1 + \sum_{l=1}^{K-1} e^{a_l^T x_k + b_l} \right)$$

is concave. Discriminant functions:  $\delta_l : \mathbf{R}^n \rightarrow \mathbf{R}$  defined by  $\delta_l(x) = p_l(a_1^T x + b_1, \dots, a_{K-1}^T x + b_{K-1})$ .

# Separating Hyperplane

Given pairs of data  $(y_k, x_k)$ ,  $k \in \mathbf{N}_N$  with  $x_k \in \mathbf{R}^n$  and  $y_k \in \{-1, 1\}$ .

Find function  $f : \mathbf{R}^n \rightarrow \mathbf{R}$  such that  $f(x_k) > 0$  if  $y_k = 1$  and  $f(x_k) < 0$  if  $y_k = -1$ .

The set  $\{x \mid f(x) = 0\}$  separates the two classes.

Let  $f : \mathbf{R}^n \times \mathbf{R}^n \times \mathbf{R} \rightarrow \mathbf{R}$ :

$$f(x; a, b) = a^T x + b$$

Objective: choose  $(a, b)$ :

$$y_k f(x_k; a, b) > 0, \quad k \in \mathbf{N}_N$$

Feasibility problem which has solution iff the two classes can be separated with the *hyperplane*  $\{x \mid a^T x + b = 0\}$ .

# Rosenblatt's Perceptron

Inequalities homogeneous in  $(a, b)$ . Equivalent to consider feasibility problem

$$y_k f(x_k; a, b) \geq 1, \quad k \in \mathbf{N}_N$$

Let  $g : \mathbf{R} \rightarrow \mathbf{R}_+$ :

$$g(y) = \begin{cases} y, & y \geq 0 \\ 0, & y < 0 \end{cases}$$

and define equivalent optimization problem

$$\text{minimize}_{a,b} \sum_{k=1}^N g(1 - y_k f(x_k; a, b))$$

Has a solution with objective function value zero iff the two classes can be separated by a hyperplane. However, not unique. Also non-differentiable objective function.

# Unique Separating Hyperplane

Maximize the distance from  $x_k$  to the hyperplane.

Distance given by  $y_k f(x_k; a, b) / \|a\|_2$ , and since  $f$  is homogeneous in  $(a, b)$  maximizing distance equivalent to minimizing  $\|a\|_2^2$ :

$$\text{minimize}_{a,b} \frac{1}{2} \|a\|_2^2 \quad (85)$$

$$\text{subject to } y_k f(x_k; a, b) \geq 1, \quad k \in \mathbf{N}_N \quad (86)$$

However, no solution if separating hyperplane not exists.

Rosenblat's perceptron provides solution anyway.

# Differentiable Perceptron Formulation

Let  $\xi = (\xi_1, \dots, \xi_N) \in \mathbf{R}^N$ , and consider

$$\text{minimize}_{a,b,\xi} \sum_{k=1}^N \xi_k \quad (87)$$

$$\text{subject to } y_k f(x_k; a, b) \geq 1 - \xi_k, \quad k \in \mathbf{N}_N \quad (88)$$

$$\xi_k \geq 0, \quad k \in \mathbf{N}_N \quad (89)$$

However, possible non-uniqueness of solution.

# Support Vector Machine (SVM)

Remedy:

$$\text{minimize}_{a,b,\xi} \sum_{k=1}^N \xi_k + \frac{\nu}{2} \|a\|_2^2 \quad (90)$$

$$\text{subject to } y_k f(x_k; a, b) \geq 1 - \xi_k, \quad k \in \mathbf{N}_N \quad (91)$$

$$\xi_k \geq 0, \quad k \in \mathbf{N}_N \quad (92)$$

where  $\nu \geq 0$  provides tradeoff.

Remark: May use regressor  $f$  nonlinear in  $x_k$ , i.e.

$f : \mathbf{R}^m \times \mathbf{R}^n \times \mathbf{R} \rightarrow \mathbf{R}$  given by

$$f(x; a, b) = a^T \beta(x) + b$$

where  $\beta : \mathbf{R}^m \rightarrow \mathbf{R}^n$



# Duality

Lagrangian  $L : \mathbf{R}^n \times \mathbf{R} \times \mathbf{R}^N \times \mathbf{R}^N \times \mathbf{R}^N \rightarrow \mathbf{R}$  for (90–92) is given by

$$L(a, b, \xi, \lambda, \mu) = \sum_{k=1}^N \xi_k + \frac{\nu}{2} \|a\|_2^2 \quad (93)$$

$$+ \sum_{k=1}^N \lambda_k (1 - \xi_k - y_k f(x_k; a, b)) - \sum_{k=1}^N \mu_k \xi_k \quad (94)$$

$$= \frac{\nu}{2} \|a\|_2^2 - \sum_{k=1}^N \lambda_k y_k a^T \beta(x_k) - \sum_{k=1}^N \lambda_k y_k b \quad (95)$$

$$+ \sum_{k=1}^N (1 - \lambda_k - \mu_k) \xi_k + \sum_{k=1}^N \lambda_k \quad (96)$$

# Minimization of Lagrangian

Minimum of Lagrangian is unbounded from below unless  $1 - \lambda_k - \mu_k = 0$ ,  $k \in \mathbf{N}_N$  and  $\sum_{k=1}^N \lambda_k y_k = 0$ . When bounded minimizing  $a$  solution of

$$\frac{\partial L}{\partial a} = \nu a - \sum_{k=1}^N \lambda_k y_k \beta(x_k) = 0$$

i.e.

$$a = \frac{1}{\nu} \sum_{k=1}^N \lambda_k y_k \beta(x_k)$$

If (91) satisfied strictly at optimality for index  $k$  complementary slackness implies  $\lambda_k = 0$ . Hence optimal solution  $a$  only depends on the  $x_k$  for which the constraint is satisfied with equality, called *support vectors*.

## Dual Problem

Lagrange dual function  $g : \mathbf{R}^N \times \mathbf{R}^N \rightarrow \mathbf{R}$ :

$$g(\lambda, \mu) = \sum_{k=1}^N \lambda_k - \frac{1}{2\nu} \sum_{k=1}^N \sum_{l=1}^N \lambda_k \lambda_l y_k y_l \beta^T(x_k) \beta(x_l)$$

with domain  $\{(\lambda, \mu) | 1 - \lambda_k - \mu_k = 0, k \in \mathbf{N}_N, \sum_{k=1}^N \lambda_k y_k = 0\}$ .

Slater's conditions fulfilled and we obtain optimal  $(\lambda, \mu)$  from

$$\text{maximize}_{\lambda, \mu} g(\lambda, \mu) \quad (97)$$

$$\text{subject to } \lambda_k + \mu_k = 1, \quad k \in \mathbf{N}_N \quad (98)$$

$$\sum_{k=1}^N \lambda_k y_k = 0 \quad (99)$$

$$\lambda_k \geq 0; \quad \mu_k \geq 0, \quad k \in \mathbf{N}_N \quad (100)$$

The so-called *sequential minimal optimization* algorithm extremely efficient.

## Solution to Primal Problem from Dual

We already have  $a$  expressed in dual variables.

For any  $\lambda_k > 0$ , complementary slackness implies  $1 - \xi_k - y_k f(x_k; a, b) = 0$ , i.e.

$$b = \frac{-\xi_k + 1}{y_k} - a^T \beta(x_k), \quad k \in \mathcal{I}$$

where  $\mathcal{I} = \{k \in \mathbf{N}_N | \lambda_k > 0\}$ .

If  $\mu_k > 0$ , complementary slackness  $\mu_k \xi_k = 0 \Rightarrow \xi_k = 0$ . Hence there is a nonempty set  $\mathcal{J} = \{k \in \mathcal{I} | \xi_k = 0\}$  for which

$$b = \frac{1}{y_k} - a^T \beta(x_k), \quad k \in \mathcal{J}$$

Any such  $k$  can be used to compute  $b$ .

For numerical stability take average.

## Kernel Trick Again

Dual problem depends only on  $\beta^T(x_k)\beta(x_l)$ . Also true for resulting regressor:

$$f(x; a, b) = \frac{1}{\nu} \sum_{k=1}^N \lambda_k y_k \beta^T(x_k) \beta(x) + b$$

Hence sufficient to know  $K : \mathbf{R}^m \times \mathbf{R}^m \rightarrow \mathbf{R}_+$  defined by  $K(x, \bar{x}) = \beta^T(x) \beta(\bar{x})$ .

# Classification using Boltzmann Machine

Given data  $(y_k, x_k)$ ,  $k \in \mathbf{N}_N$  with  $x_k \in \{0, 1\}^M$  and  $y_k \in \{0, 1\}^K$ .

$(y_k, x_k)$  belongs to class  $l \in \mathbf{N}_K$  if  $y_k = e_l$ , where  $e_l$  is the  $l$ th unit vector.

We let  $v_k = (y_k, x_k) \in \{0, 1\}^{M+N}$  be the visible variables in the Boltzmann machine.

Use data  $(y_k, x_k)$ ,  $k \in \mathbf{N}_N$  to find optimal values of the parameters  $(\lambda, \Lambda)$  in Ising distribution  $p(y, x, h)$ , where  $h$  are the hidden variables

Let  $\delta_l(x) = \sum_h p(e_l, x, h)$  be discriminant function and say that  $x$  belongs to class  $l$  if  $\delta_l(x) > \delta_k(x)$  for all  $k \neq l$ .

# Restricted Boltzmann Machine (RBM)

Graphical Ising model which is obtained by considering a graph on the variables  $(v, h)$  such that there are no edges between the variables within  $v$  or  $h$ , respectively. Then

$$\Lambda = \begin{bmatrix} 0 & \Lambda_{12} \\ \Lambda_{12}^T & 0 \end{bmatrix}$$

which means Ising distribution reads:

$$p(v, h) = \frac{1}{Z} e^{\lambda_1^T v + \lambda_2^T h + 2v^T \Lambda_{12} h}$$

where  $Z$  normalizes  $p$  to a distribution.

# Factorization of Conditional Distribution

Marginal distribution of  $v$ :

$$\begin{aligned}\sum_h p(v, h) &= \frac{e^{\lambda_1^T v}}{Z} \sum_h e^{(\lambda_2 + 2\Lambda_{12}^T v)^T h} = \frac{e^{\lambda_1^T v}}{Z} \sum_h \prod_i e^{(\lambda_2 + 2\Lambda_{12}^T v)_i^T h_i} \\ &= \frac{e^{\lambda_1^T v}}{Z} \prod_i \sum_{h_i} e^{(\lambda_2 + 2\Lambda_{12}^T v)_i^T h_i} = \frac{e^{\lambda_1^T v}}{Z} \prod_i \left(1 + e^{(\lambda_2 + 2\Lambda_{12}^T v)_i^T}\right)\end{aligned}$$

Conditional distribution:

$$\frac{p(v, h)}{\sum_h p(v, h)} = \frac{\frac{1}{Z} e^{\lambda_1^T v + \lambda_2^T h + 2v^T \Lambda_{12} h}}{\frac{e^{\lambda_1^T v}}{Z} \prod_i \left(1 + e^{(\lambda_2 + 2\Lambda_{12}^T v)_i^T}\right)} = \frac{\prod_i e^{(\lambda_2 + 2\Lambda_{12}^T v)_i^T h_i}}{\prod_i \left(1 + e^{(\lambda_2 + 2\Lambda_{12}^T v)_i^T}\right)}$$

Conditional distribution for  $h_j$  given  $v$ :

$$s_j(h_j, v) = \frac{e^{(\lambda_2 + 2\Lambda_{12}^T v)_j^T h_j}}{\prod_i \left(1 + e^{(\lambda_2 + 2\Lambda_{12}^T v)_i^T}\right)}$$



# Logistic Function

We have

$$s_j(0, \nu) = \frac{1}{\prod_i (1 + e^{x_i})}; \quad s_j(1, \nu) = \frac{e^{x_j}}{\prod_i (1 + e^{x_i})}$$

respectively, where  $x_i = (\lambda_2 + 2\Lambda_{12}^T \nu)_i$ . Since  $s_j(0, \nu) + s_j(1, \nu) = 1$ , we have that  $\prod_i (1 + e^{x_i}) = e^{x_j} + 1$ , and hence

$$s_j(1, \nu) = \frac{e^{x_j}}{1 + e^{x_j}} = \frac{1}{1 + e^{-x_j}} = \sigma(x_j)$$

where  $\sigma : \mathbf{R} \rightarrow \mathbf{R}$  is the *logistic function*, which is an example of a so-called *sigmoid function*.

# Gradient Expressions

Assume only one observation  $v_k$ , which we write as  $v$ . Case of several observations is obtained by summing up the gradients for the different observations.<sup>1</sup>

Gradients:

$$\frac{\partial Q}{\partial \lambda} = \sum_h s(h, v, \lambda^-, \Lambda^-) \begin{bmatrix} v \\ h \end{bmatrix} - \sum_{\xi_v} p_v(\xi_v) \sum_{\xi_h} s(\xi_h, \xi_v, \lambda^-, \Lambda^-) \begin{bmatrix} \xi_v \\ \xi_h \end{bmatrix}$$
$$\frac{\partial Q}{\partial \Lambda_{12}} = 2 \sum_h s(h, v, \lambda^-, \Lambda^-) v h^T - \sum_{\xi_v} p_v(\xi_v) \sum_{\xi_h} s(\xi_h, \xi_v, \lambda^-, \Lambda^-) \xi_v \xi_h^T$$

where  $s(h, v, \lambda^-, \Lambda^-) = \prod_k s_k(h_k, v)$ ,  $\xi = (\xi_v, \xi_h)$ , and where  $p_v(\xi_v) = \sum_{\xi_h} p(\xi_v, \xi_h)$  marginal distribution for the observations.

---

<sup>1</sup>Sub-indexes will from now on refer to component of vectors.

## Simplifications

We have:

$$\begin{aligned}\sum_h s(h, v, \lambda^-, \Lambda^-) h_l &= \sum_h \prod_k s_k(h_k, v) h_l = \sum_h \prod_{k \neq l} s_k(h_k, v) s_l(h_l, v) h_l \\ &= \prod_{k \neq l} \sum_{h_k} s_k(h_k, v) \sum_{h_l} s_l(h_l, v) h_l = s_l(1, v)\end{aligned}$$

where the last equality follows from the fact that  $s_k(0, v) + s_k(1, v) = 1$  and  $s_l(0, v) \times 0 = 0$ , and hence

$$\begin{aligned}\frac{\partial Q}{\partial \lambda_1} &= v - \sum_{\xi_v} p_v(\xi_v) \xi_v \\ \left( \frac{\partial Q}{\partial \lambda_2} \right)_i &= s_i(1, v) - \sum_{\xi_v} p_v(\xi_v) s_i(1, \xi_v) \\ \left( \frac{\partial Q}{\partial \Lambda_{12}} \right)_{i,j} &= 2s_j(1, v) v_i - \sum_{\xi_v} p_v(\xi_v) s_j(1, \xi_v) (\xi_v)_i\end{aligned}$$

# Contrastive Divergence Method

Approximated second term in gradients using Gibbs sampling from  $p_v$ .

1. Initialize Gibbs sampler with  $\xi_v^{(0)} = v$ .
2. Because of graphical structure of RBM we then draw a sample  $h^{(1)}$  from the conditional distribution of  $h$  given  $v$ , i.e. from  $s_j(h_j, \xi_v^{(0)})$ .
3. Draw a sample  $\xi_v^{(1)}$  from the conditional distribution of  $v$  given  $h$ , where we use  $h = h^{(1)}$ .<sup>2</sup>

Approximation of sums for gradients:

$$\begin{aligned} & \sum_{\xi_v} \xi_v^{(1)} \\ & \sum_{\xi_v} s_i(1, \xi_v^{(1)}) \\ & \sum_{\xi_v} s_j(1, \xi_v^{(1)}) (\xi_v^{(1)})_i \end{aligned}$$

---

<sup>2</sup>This conditional distribution can be obtained similarly to  $s_j(h_j, v)$  above. 

# Artificial Neural Networks (ANN)s

An ANN with  $L$  layers and  $n_l$  neurons for layer  $l$  is a nonlinear regression model.

Let  $a(l) \in \mathbf{R}^{n_l}$  for  $0 \leq l \leq L - 1$  be the *input activation* of layer  $l$ , and let

$$z(l) = \Phi(l; a(l-1)), \quad l \in \mathbf{N}_L \quad (101)$$

where  $z(l) \in \mathbf{R}^{n_l}$  is the *output* of layer  $l$ , and where  $\Phi(l; \cdot) : \mathbf{R}^{n_{l-1}} \rightarrow \mathbf{R}^{n_l}$  is called the *propagation function*.

Input to the next layer:

$$a(l) = h(l; z(l)), \quad l \in \mathbf{N}_{L-1} \quad (102)$$

where  $h(l; \cdot) : \mathbf{R}^{n_l} \rightarrow \mathbf{R}^{n_l}$  is called the *activation function*.

## Special Cases

If  $\Phi$  affine, we may write

$$z(l) = W(l)a(l-1) + v(l), \quad l \in \mathbf{N}_L \quad (103)$$

where  $W(l) \in \mathbf{R}^{n_l \times n_{l-1}}$  and  $v(l) \in \mathbf{R}^{n_l}$ .

It is often the case that

$$h(l; z(l)) = (h_1(l; z_1(l)), \dots, h_{n_l}(l; z_{n_l}(l)))$$

with  $h_i$  typically a saturation function or a sigmoid function, i.e. a function  $\sigma : \mathbf{R} \rightarrow [0, 1]$  such that  $\lim_{t \rightarrow \infty} \sigma(t) = 1$  and  $\lim_{t \rightarrow -\infty} \sigma(t) = 0$ .

## Representation and Approximation Capability

By Kolmogorov (1957) and Lorentz (1976) any continuous function  $f : [0, 1]^n \rightarrow \mathbf{R}$  can be represented as

$$f(x) = \sum_{q=0}^{2n} g \left( \sum_{p=1}^n \phi_{p,q}(x_p) \right) \quad (104)$$

where  $\phi_{p,q} : [0, 1] \rightarrow [0, 1]$  are continuous increasing functions, and where  $g : \mathbf{R} \rightarrow \mathbf{R}$  is a continuous function, Hence a two layer ANN is sufficient.

By Cybenko (1989) any continuous function  $f : [0, 1]^n \rightarrow \mathbf{R}$  can be approximated arbitrarily well with

$$f(x) = \sum_{j=1}^N \alpha_j \sigma(a_j^T x + b_j) \quad (105)$$

with any continuous sigmoid function, where  $\alpha_j \in \mathbf{R}$ ,  $a_j \in \mathbf{R}^n$  and  $b_j \in \mathbf{R}$ ,  $j \in \mathbf{N}_N$ .

## Regression with ANN

For a given sequence of input activations  $a(0) = (a^1(0), \dots, a^N(0))$  select ANN that reproduces as accurately as possible a given output sequence  $y = (y^1, \dots, y^N)$  of layer  $L$ , i.e.  $y^k$  and  $z^k(L)$  should be close to one another for all  $k \in \mathbf{N}_N$ .

Closeness measured with function  $V : \mathbf{R}^{2Nn_L} \rightarrow \mathbf{R}$

Example:  $V(z(L), y) = \frac{1}{2} \|z(L) - y\|_2^2$ , i.e. a nonlinear LS problem.

Superscript refers to a particular data set, and no superscript denotes the vector of all data sets and corresponding variables.

How to select the ANN?

With linear or affine equation for the propagation function we may optimize  $V$  with respect to  $W(l)$  and  $v(l)$ . We assume that the functions  $h(l; \cdot)$  are given and fixed.



# Optimization Problem

$$\text{minimize}_{W, v} V(z(L), y) \quad (106)$$

where  $W = (W(1), \dots, W(L))$  and  $v = (v(1), \dots, v(L))$ , where we with abuse of notation consider  $V$  to be a function of  $(W, v)$ , wick it is by recursive substitution (102–103).

Typically we have that  $V(z(L), y) = \sum_{k=1}^N V^k(z^k(L), y^k)$ , i.e. we have a sum of functions for each data set which share the same parameters  $W$  and  $v$  for the ANN. True for nonlinear LS above.

We may also consider a constrained version of the problem above by also considering  $z(l)$  and  $a(l)$ ,  $l \in \mathbf{N}_{L-1}$  as variables and (102–103) as constraints.

# Interpretation of Logistic Regression as ANN

Let  $a(0) = x$ ,

$$W(1) = \begin{bmatrix} a_1^T \\ \vdots \\ a_{K-1}^T \end{bmatrix}; \quad v(0) = \begin{bmatrix} b_1 \\ \vdots \\ b_{K-1} \end{bmatrix}$$

and  $h(z) = (z(1)/s, z(2)/s, \dots, 1/s)$ , where  $s = 1 + \sum_{k=1}^{K-1} e^{z(k)}$ .

Hence output  $a(1)$  of the ANN are the probabilities  $p_l(z_1, z_2, \dots, z_{K-1})$ ,  $l \in \mathbf{N}_K$  of the categorical distribution, i.e.  $a(1)_l = p_l(z_1, z_2, \dots, z_{K-1})$ .

The training data is  $(x^k, y^k) \in \mathbf{R}^n \times \{0, 1\}$  for  $k \in \mathbf{N}_N$

Function to maximize is the likelihood function for the data.

# Interpretation of RBM as ANN

Let  $a(0) = v$ ,  $v(0) = \lambda_2$ ,  $W(1) = 2\Lambda_{12}^T$  and  $h_i(z_i(1)) = \sigma(z_i(1))$ .

Then  $a(1)_i = s_i(1, v)$ , and hence the outputs of the ANN are the conditional probabilities for  $h_i = 1$  given  $v$ .

Training data is  $v = (v^1, \dots, v^N)$  for the visible layer of the RBM

Function to maximize is the likelihood function for the data.

# Interpretation of Perceptron as ANN

Let  $a(0) = x$ ,  $v(0) = b$ ,  $W(1) = a^T$  and  $h(z(1)) = g(z(1))$ .

Training data is  $(x_k, y_k) \in \mathbf{R}^n \times \{-1, 1\}$ ,  $k \in \mathbf{N}_N$

Function to maximize is the sum of the outputs of the ANN over all training data.

# Consequence

Because of these interpretations it easy to see how one can generalize logistic regression, the RBM, and the perceptron using multi-layer ANNs.

## Least Squares (LS) Problem

$$\text{minimize}_{x \in \mathbf{R}^n} \frac{1}{2} \sum_{k=1}^m (a_k(x) - b_k)^2 \quad (107)$$

where  $a_k : \mathbf{R}^n \rightarrow \mathbf{R}$  and  $b_k \in \mathbf{R}$ .

Compactly:

$$\text{minimize}_{x \in \mathbf{R}^n} \frac{1}{2} \|a(x) - b\|_2^2$$

where  $\|\cdot\|_2$  is the Euclidian norm and where  $a(x) = (a_1(x), \dots, a_m(x))$  and  $b = (b_1, \dots, b_m)$ .

Linear LS:  $a_k(x) = a_k^T x$ , where with abuse of notation  $a_k \in \mathbf{R}^n$ .

For the linear case  $a(x) = Ax$ , where  $A = \begin{bmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_m^T \end{bmatrix} \in \mathbf{R}^{m \times n}$

# Optimality Conditions

Necessary conditions for optimality of nonlinear LS problem:

$$\frac{\partial}{\partial x} \frac{1}{2} \sum_{k=1}^m (a_k(x) - b_k)^2 = \sum_{k=1}^m (a_k(x) - b_k) \frac{\partial a_k(x)}{\partial x} = 0 \quad (108)$$

For linear case optimality conditions are:

$$\sum_{k=1}^m (a_k^T x - b_k) a_k = 0 \quad (109)$$

or equivalently

$$A^T A x = A^T b$$

called *Normal equation*.

# Constrained LS Problem

For linear case:

$$\text{minimize}_{x \in \mathbf{R}^n} \frac{1}{2} \|Ax - b\|_2^2 \quad (110)$$

$$\text{subject to } Bx = c \quad (111)$$

where  $B \in \mathbf{R}^{p \times n}$ .

Optimality conditions (KKT equation):

$$\begin{bmatrix} A^T A & B^T \\ B & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} A^T b \\ c \end{bmatrix} \quad (112)$$

where  $\lambda \in \mathbf{R}^p$

Constraint qualification is satisfied if  $B$  full row rank.

Unique solution if and only if  $A^T A$  is positive definite on the nullspace of  $B$ .



# Quadratic Program (QP)

$$\text{minimize}_{x \in \mathbf{R}^n} \frac{1}{2} x^T Q x + r^T x \quad (113)$$

$$\text{subject to } Bx = c \quad (114)$$

where  $Q \in \mathbf{S}_+^n$ ,  $r \in \mathbf{R}^n$ ,  $B \in \mathbf{R}^{p \times n}$ , and  $c \in \mathbf{R}^p$ .

LS problem is a QP if  $Q = A^T A$  and  $r = A^T b$ .

## QP with Affine Inequality Constraints

$$\text{minimize}_{x \in \mathbf{R}^n} \frac{1}{2} x^T Q x + r^T x \quad (115)$$

$$\text{subject to } Bx = c \quad (116)$$

$$Cx \leq d \quad (117)$$

where  $C \in \mathbf{R}^{q \times m}$  and  $d \in \mathbf{R}^m$ .

KKT optimality conditions:

$$Qx + B^T \mu + C^T \lambda = -r$$

$$Bx - c = 0$$

$$Cx + s - d = 0$$

$$s_i \lambda_i = 0, \quad i \in \mathbf{N}_q$$

together with  $\lambda \geq 0$  and  $s \geq 0$ , where  $\mu \in \mathbf{R}^p$ ,  $\lambda \in \mathbf{R}^q$ .

Special case when  $Q = 0$ , which is called a *Linear Program* (LP).

# Cones

A set  $C$  is a *cone* if for any  $x \in C$  and for any  $\theta \geq 0$  it holds that  $\theta x \in C$ .

A set  $C$  is a *convex cone* if it is convex and a cone.

This holds if for any  $x, y \in C$  and any  $\theta_1, \theta_2 \geq 0$  it holds that  $\theta_1 x + \theta_2 y \in C$ .

A cone is *proper* if it is closed, pointed, i.e.  $C \cap (-C) = \{0\}$  and has nonempty interior.

# Examples of Cones

*First orthant:*

$$\mathbf{R}_+^n = \{x \mid x \geq 0\} \subseteq \mathbf{R}^n$$

*Norm cone:*

$$\{(x, t) \in \mathbf{R}^{n-1} \times \mathbf{R} \mid \|x\| \leq t\} \subseteq \mathbf{R}^n$$

where  $\|\cdot\| : \mathbf{R}^{n-1} \rightarrow \mathbf{R}$  is any norm.

When the norm is the Euclidian norm we get the *Second-order cone*, the *quadratic cone*, the *Lorentz cone* or the *ice-cream cone* denoted  $\mathbf{Q}^n$ .

*Positive semidefinite cone:*

$$\mathbf{S}_+^n = \{X \in \mathbf{S}^m \mid X \succeq 0\}$$

# Epigraph of Perspective of Convex Function

Let  $f : \mathbf{R}^n \rightarrow \mathbf{R}$  be a convex function and

$$K = \{(x, y, z) \mid y > 0, yf(x/y) \leq z\}$$

Then  $K \cup \{0, 0, 0\}$  is a convex cone.

If the interior of **dom**  $f$  is not empty and if **epi**  $f$  does not contain a straight line, then the cone is proper.

Any convex inequality constraint  $f(x) \leq t$  can be rephrased as convex conic constraint  $(x, 1, t) \in K$  modulo some minor modifications to make  $K$  proper.

*Exponential cone* when  $f(x) = \exp(x)$  and  $n = 1$ :

$$K_{\text{exp}} = \{(x, y, z) \in \mathbf{R}^3 \mid y > 0, y \exp(x/y) \leq z\} \cup \{(0, 0, 0)\}$$

# Conic Optimization Problem

The following optimization problem is called a *conic optimization problem*:

$$\text{minimize}_x c^T x \quad (118)$$

$$\text{subject to } Ax = b \quad (119)$$

$$x \in C \quad (120)$$

where  $A \in \mathbf{R}^{p \times n}$ ,  $b \in \mathbf{R}^p$ ,  $c \in \mathbf{R}^n$  and  $C \subseteq \mathbf{R}^n$  is a proper cone.

There are special purpose solver for conic optimization problems, e.g. MOSEK.

# Dual Cone

$$C^* = \{y \mid \langle x, y \rangle \geq 0 \ \forall x \in C\}$$

where  $\langle \cdot, \cdot \rangle$  is appropriate inner product.

If  $C$  proper cone, then  $C^*$  proper cone,  $(C^*)^* = C$  and

$$\text{int}C^* = \{y \mid \langle x, y \rangle > 0 \ \forall x \in C, x \neq 0\}$$

A cone is self-dual if  $C^* = C$ . The cones  $\mathbf{R}_+^n$ ,  $\mathbf{Q}^n$ , and  $\mathbf{S}_+^n$  are self-dual.

The dual of the exponential cone  $K_{\text{exp}}$  is

$$K_{\text{exp}}^* = \{(u, v, w) \in \mathbf{R}_- \times \mathbf{R} \times \mathbf{R}_+ \mid -u \ln(-u/w) + u - v \leq 0\}$$

# Duality

Lagrangian  $L : \mathbf{R}^n \times \mathbf{R}^n \times \mathbf{R}^p \rightarrow \mathbf{R}$  for conic optimization problem:

$$L(x, \lambda, \mu) = c^T x - \lambda^T x + \mu^T (Ax - b)$$

Assume that  $C \subseteq \mathbf{R}^n$ .<sup>3</sup>

Lagrange dual function  $g : \mathbf{R}^n \times \mathbf{R}^p \rightarrow \mathbf{R}$ , where

$$g(\lambda, \mu) = \min_x L(x, \lambda, \mu)$$

where  $g(\lambda, \mu) = -b^T \mu$  when  $c - \lambda - A^T \mu = 0$  and  $-\infty$  otherwise.

$g(\lambda, \mu) \leq p^*$  for  $\lambda \in C^*$ .

---

<sup>3</sup>For  $\mathbf{S}_+^m$  we use a symmetric vectorization of  $X \in \mathbf{S}^m$ , i.e.  $x = \mathbf{svec} X$ , where  $\mathbf{svec} : \mathbf{S}^m \rightarrow \mathbf{R}^n$  with  $n = m(m-1)/2$ .



# Lagrange Dual Problem

$$\text{maximize}_{\lambda, \mu} -b^T \mu \quad (121)$$

$$\text{subject to } A^T \mu + \lambda = c \quad (122)$$

$$\lambda \in C^* \quad (123)$$

KKT conditions:

$$A^T \mu + \lambda = c$$

$$Ax = b$$

$$x^T \lambda = 0$$

together with  $x \in C$  and  $\lambda \in C^*$ .

Necessary and sufficient for optimality if either the primal problem or the dual problem admits a strictly feasible point.

# Entropy Maximization as Conic Problem

$(x, y, z) \in K_{\text{exp}}$  implies  $z \geq 0$ , and hence equivalently  $x \leq y \ln(z/y)$  and  $(y, z) \in \mathbf{R}_+^2$ .

Entropy has terms  $-p_k \ln p_k$ , and  $t_k \leq -p_k \ln p_k$  can equivalently be written  $t_k \leq p_k \ln(1/p_k)$  or  $(t_k, p_k, 1) \in E_{\text{exp}}$ .

Hence the entropy problem is equivalent to

$$\begin{aligned} & \text{minimize}_{p,t} \quad -\mathbf{1}^T t \\ & \text{subject to} \quad Ap = b \\ & \quad \quad \quad \mathbf{1}^T p = 1 \\ & \quad \quad \quad (t_k, p_k, 1) \in K_{\text{exp}}, \quad k \in \mathbf{N}_n \end{aligned}$$

where  $t = (t_1, \dots, t_n)$ .  $((t_k, p_k, 1) \in K_{\text{exp}} \Rightarrow p \in \mathbf{R}_+^n)$

# Logistic Regression as Conic Problem

$$\begin{aligned} \text{minimize}_{a,b,t,z} \quad & \sum_{k=1}^N t_k - \sum_{k:y_k=l, l \in \mathbf{N}_{K-1}} a_l^T x_k + b_l \\ & \begin{bmatrix} a_l^T x_k + b_l - t_k \\ 1 \\ z_{k,l} \end{bmatrix} \in K_{\text{exp}}, \quad k \in \mathbf{N}_N, l \in \mathbf{N}_{K-1} \\ & \begin{bmatrix} -t_k \\ 1 \\ z_{k,K} \end{bmatrix} \in K_{\text{exp}}, \quad k \in \mathbf{N}_N \\ & 1 - \sum_{l=1}^K z_{k,l} \geq 0, \quad k \in \mathbf{N}_N \end{aligned}$$

where  $a = (a_1, \dots, a_{K-1})$ ,  $b = (b_1, \dots, b_{K-1})$ , and  $t = (t_1, \dots, t_N)$  with  $t_k \in \mathbf{R}$  and  $z \in \mathbf{R}^{N \times K}$ .

# Rank -Constrained Optimization

Let  $X, Z \in \mathbf{R}^{m \times n}$ , and consider:

$$\underset{Z}{\text{minimize}} \quad \frac{1}{2} \|Z - X\| \quad (124)$$

$$\text{subject to} \quad \mathbf{Rank} \, Z \leq r \quad (125)$$

where  $r \leq q = \min(m, n)$  and where  $\|\cdot\|$  denotes the matrix norm induced by the Euclidian norm.

Let  $X = UDV^T$  be an SVD with diagonal elements  $d_i$  of  $D$  in decreasing order.

$Z = U_1 D_1 V_1^T$ , where  $U_1$  and  $V_1$  contains the first  $r$  columns of  $U$  and  $V$ , respectively, and where  $D = D_1 \oplus D_2$  is an optimal  $Z$ .

## Relation to PCA

For observations  $x_i \in \mathbf{R}^n$ ,  $i \in \mathbf{N}_N$  let

$$X = \begin{bmatrix} x_1^T \\ \vdots \\ x_N^T \end{bmatrix}$$

In PCA this matrix is approximated with  $XW^T W = U_1 D_1 V_1^T$ , which is also the solution to the rank-constrained problem

The objective function that is minimized in PCA is the Frobenius norm of  $X(I - W^T W) = X - XW^T W$ .

In the rank-constrained problem we instead minimize the induced norm of  $X - Z$ .

In the rank-constrained optimization we have an explicit constraint on the rank of  $Z$ , whereas we in principal component analysis constrain the rank implicitly with the number of rows in  $W$ , which are equal to the constraint on the rank.

# Rank Optimization

Rank-constrained optimization is closely related to the *rank optimization* problem:

$$\text{minimize}_Z \quad \mathbf{Rank} \, Z \quad (126)$$

$$\text{subject to } \frac{1}{2} \|Z - X\| \leq \varepsilon \quad (127)$$

where  $X, Z \in \mathbf{R}^{m \times n}$ , and where  $\varepsilon \in \mathbf{R}_+$ .

The solution for one of them can readily be obtained from the other by doing bi-sectioning on  $r$  or  $\varepsilon$ , respectively.

# General Rank Minimization Problem

$$\text{minimize}_Z \quad \mathbf{Rank} \, Z \quad (128)$$

$$\text{subject to } Z \in \mathcal{C} \quad (129)$$

where  $\mathcal{C} \subset \mathbf{R}^{m \times n}$  typically a convex set.

The overall problem is however not convex.

# Heuristic Rank Optimization

Assume that  $Z$  is square and positive semidefinite.

Heuristic optimization problem:

$$\text{minimize}_Z \quad \text{tr } Z \quad (130)$$

$$\text{subject to } Z \in \mathcal{C} \quad (131)$$

$$(132)$$

where  $Z \in \mathbf{S}_+^m$ .

This follows from the fact that  $\text{tr } Z = \sum_{i=1}^m \lambda_i(Z) = \|\lambda(Z)\|_1$ , where  $\lambda(Z)$  is the vector of eigenvalues of  $Z$ . Since minimizing the  $\ell_1$ -norm is known to result in a sparse vector, we will obtain several eigenvalues that are zero, and hence a low-rank matrix  $Z$ .



# General $Z$

For general  $Z \in \mathbf{R}^{m \times n}$  it holds that  $\mathbf{Rank} Z \leq r$  if and only if there exist matrices such that  $\mathbf{Rank} X + \mathbf{Rank} Y \leq 2r$  and

$$\begin{bmatrix} X & Z \\ Z^T & Y \end{bmatrix} \succeq 0$$

Because of this we consider the heuristic

$$\text{minimize}_{X,Y,Z} \frac{1}{2} \mathbf{tr} X \oplus Y \quad (133)$$

$$\text{subject to } \begin{bmatrix} X & Z \\ Z^T & Y \end{bmatrix} \succeq 0 \quad (134)$$

$$Z \in \mathcal{C} \quad (135)$$

# Nuclear Norm Interpretation

Equivalent problem:

$$\text{minimize}_Z \|Z\|_* \quad (136)$$

$$\text{subject to } Z \in \mathcal{C} \quad (137)$$

where  $\|\cdot\|_* : \mathbf{R}^{m \times n} \rightarrow \mathbf{R}_+$  is the *nuclear norm* or *Kay-Fan  $n$ -norm* of  $Z$ .

This norm is the dual of the induced norm and given by

$$\|Z\|_* = \sum_{i=1}^q \sigma_i(Z)$$

where  $q = \min(m, n)$ , and where  $\sigma_i(Z)$  are the singular values of  $Z$ .

# Convex Envelope

Can be shown that  $\|Z\|_*$  is the *convex envelope* of **Rank**  $Z$  on the set  $\mathcal{D} = \{Z \in \mathbf{R}^{m \times n} \mid \|Z\| \leq 1\}$ .

That means that  $\|Z\|_*$  is the largest convex function such that  $\|Z\|_* \leq \mathbf{Rank} \, Z$  for all  $Z \in \mathcal{D}$ .

Hence the nuclear norm heuristic replaces the nonconvex rank-function with the tightest lower convex bound.

However, it is only tight in case we consider the constraint set  $\mathcal{D}$ , but in many cases it is a good heuristic also without this constraint.

# Log-det Heuristic

For a low rank matrix  $Z \in \mathbf{S}_+^m$  and for small values of  $\delta > 0$  we have  $\ln \det(Z + \delta I)$  negative with a large absolute value.

Heuristic for rank minimization when  $Z \in \mathbf{S}_+^m$ :

$$\text{minimize}_Z \ln \det(Z + \delta I) \quad (138)$$

$$\text{subject to } Z \in \mathcal{C} \quad (139)$$

Notice that  $\ln \det(Z + \delta I)$  is not a convex function of  $Z$  but instead concave.

# General $Z$

For general  $Z \in \mathbf{R}^{m \times n}$  we just as before embed the problem and obtain the heuristic

$$\text{minimize}_{X,Y,Z} \ln \det(X \oplus Y + \delta I) \quad (140)$$

$$\text{subject to } \begin{bmatrix} X & Z \\ Z^T & Y \end{bmatrix} \succeq 0 \quad (141)$$

$$Z \in \mathcal{C} \quad (142)$$

where  $X \in \mathbf{S}^m$  and  $Y \in \mathbf{S}^n$ .

# Gradient Descent Method

Consider

$$\text{minimize } f(x) \quad (143)$$

where  $f: \mathbf{R}^n \rightarrow \mathbf{R}$  is continuously differentiable, i.e., gradient of  $f(x)$

$$\nabla f(x) = \begin{bmatrix} \partial f / \partial x_1 \\ \vdots \\ \partial f / \partial x_n \end{bmatrix}$$

exists and is a continuous function.

Assume  $f$  bounded below and that minimum attained. Given initial point  $x^0$ , *gradient descent method* updates

$$x^{k+1} = x^k - t_k \nabla f(x^k) \quad (144)$$

where  $t_k > 0$  is the step size.

## $\beta$ -smoothness

The function  $f$  is said to be  $\beta$ -smooth if  $\exists \beta > 0$ :

$$f(y) \leq f(x) + \nabla f(x)^T(y - x) + \frac{\beta}{2}\|y - x\|_2^2, \quad \forall x, y \in \mathbf{R}^n. \quad (145)$$



$$\|\nabla f(y) - \nabla f(x)\|_2 \leq \beta\|y - x\|_2, \quad \forall x, y \in \mathbf{R}^n. \quad (146)$$

For twice differentiable  $f$ ,  $\beta$ -smoothness implies

$$\|\nabla^2 f(x)\|_2 \leq \beta, \quad \forall x.$$

# Majorization Minimization Principle

Let  $\bar{f}(\cdot | \cdot) : \mathbf{R}^n \times \mathbf{R}^n \rightarrow \mathbf{R}$  be given by

$$\bar{f}(y | x) = f(x) + \nabla f(x)^T (y - x) + \frac{\beta}{2} \|y - x\|_2^2$$

Then  $f(y) \leq \bar{f}(y | x)$  and  $\bar{f}(x | x) = f(x)$  which implies that  $\bar{f}$  majorizes  $f$  at  $x$ .

The *majorization minimization algorithm* is

$$x^{k+1} = \underset{y}{\operatorname{argmin}} \bar{f}(y | x^k), \quad k = 0, 1, 2, \dots, \quad (147)$$

where  $x^0 \in \mathbf{R}^n$  given.

Descent method, i.e.  $f(x^{k+1}) \leq f(x^k)$ .



# Interpretation as Gradient Descent

We have

$$\operatorname{argmin}_y \bar{f}(y | x) = x - \frac{1}{\beta} \nabla f(x),$$

and hence (147) is gradient descent

$$x^{k+1} = x^k - \frac{1}{\beta} \nabla f(x^k), \quad k = 0, 1, 2, \dots \quad (148)$$

with step size  $t_k = 1/\beta$ .

# Convergence

Substituting  $x^{k+1}$  for  $y$  and  $x^k$  for  $x$  in (145), we obtain

$$f(x^{k+1}) \leq f(x^k) - \frac{1}{2\beta} \|\nabla f(x^k)\|_2^2$$

which shows that (148) always reduces the objective value unless  $x^k$  is a stationary point.

If  $f$  is bounded below

$$\lim_{k \rightarrow \infty} \|\nabla f(x^k)\|_2 = 0$$

If  $f$  is convex, then convergence to global optimum.

## Convergence ctd.

If  $f$  is differentiable, convex, and  $\beta$ -smooth, then

$$f(x^{(k)}) - p^* = O(1/k)$$

where  $p^*$  is optimal value, i.e. *sublinear* convergence rate.

If  $f$  is also  $\alpha$ -strongly convex, then

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{\alpha}{2} \|y - x\|_2^2, \quad \forall x, y \in \mathbf{R}^n,$$

for some  $\alpha > 0$ . With constant  $t_k = t$ :

$$\|x^k - x^*\|_2^2 \leq \left(1 - \frac{2t\alpha\beta}{\alpha + \beta}\right)^k \|x^0 - x^*\|_2^2$$

where  $0 < t \leq 2/(\alpha + \beta)$ . Best upper bound for  $t = 2/(\alpha + \beta)$ :

$$1 - \frac{2t\alpha\beta}{\alpha + \beta} = \left(\frac{\beta - \alpha}{\beta + \alpha}\right)^2 = \left(\frac{\kappa - 1}{\kappa + 1}\right)^2$$

where  $\kappa = \beta/\alpha$ , If  $f$  in addition twice continuously differentiable, then the eigenvalues  $\lambda(\nabla^2 f(x)) \in [\alpha, \beta]$ .

# Proximal Gradient Method

Consider

$$\text{minimize } g(x) + h(x) \quad (149)$$

where  $g$  is continuously differentiable and  $\beta$ -smooth, and  $h$  is proper, closed, and convex but not necessarily differentiable. Then

$$\bar{f}(y | x) = g(x) + \nabla g(x)^T (y - x) + \frac{\beta}{2} \|y - x\|_2^2 + h(y)$$

is a majorization of  $f = g + h$  at  $x$ .

Majorization minimization algorithm results in *proximal gradient method*:

$$\begin{aligned} x^{k+1} &= \underset{y}{\operatorname{argmin}} \bar{f}(y | x^k) \\ &= \underset{y}{\operatorname{argmin}} \left\{ h(y) + \nabla g(x^k)^T y + \frac{\beta}{2} \|y - x^k\|_2^2 \right\} \\ &= \underset{y}{\operatorname{argmin}} \left\{ h(y) + \frac{\beta}{2} \|y - (x^k - \frac{1}{\beta} \nabla g(x^k))\|_2^2 \right\}. \end{aligned}$$

# Proximal Operator

Equivalent formulation:

$$x^{k+1} = \mathbf{prox}_{\beta^{-1}h} \left( x^k - \beta^{-1} \nabla g(x^k) \right) \quad (150)$$

where

$$\mathbf{prox}_h(x) = \underset{y}{\operatorname{argmin}} \left\{ h(y) + \frac{1}{2} \|y - x\|_2^2 \right\} \quad (151)$$

is so-called *proximal operator* associated with the function  $h$ .

## Optimality Conditions

First-order necessary optimality condition for (149) implies that a stationary point  $x^*$  must satisfy

$$-\nabla g(x^*) \in \partial h(x^*) \quad (152)$$

where

$$\partial h(x) = \{v \in \mathbf{R}^n \mid h(y) \geq h(x) + v^T(y - x) \text{ for all } y \in \mathbf{R}^n\} \quad (153)$$

is the *subdifferential* of  $h$ .<sup>4</sup>

The stationarity condition for the majorization  $\bar{f}(y \mid x^*)$ , i.e.,

$$-\nabla g(x^*) - \beta(y - x^*) \in \partial h(y),$$

agree with optimality conditions when  $y = x^*$ .

Hence a fixed-point of the iteration (150) is a stationary point  $x^*$  of  $f = g + h$ .

---

<sup>4</sup>If  $h$  is differentiable at  $x$ , then the subdifferential of  $h$  at  $x$  is the singleton  $\{\nabla h(x)\}$ .

# Convergence Rate

When both  $g$  and  $h$  are convex, the proximal gradient iteration satisfies the bound

$$f(x^{(k)}) - p^* = O(1/k).$$

An overview of other variants of the proximal gradient method, including a detailed analysis of both the convex and the nonconvex case, can be found in Beck [2017].

# Constrained Optimization Problems

Consider

$$\begin{array}{ll} \text{minimize} & g(x) \\ \text{subject to} & x \in C \end{array} \quad (154)$$

where  $C$  is a convex subset of  $\mathbf{R}^n$ .

Problem equivalent to (149) if we define  $h(x) = \delta_C(x)$  where  $\delta_C$  indicator function of set  $C$  defined as

$$\delta_C(x) = \begin{cases} 0, & x \in C \\ \infty, & x \notin C. \end{cases}$$

In this special case, proximal operator  $\mathbf{prox}_h(x)$  becomes Euclidean projection of  $x$  onto  $C$ , and resulting method is referred to as the *projected gradient* or *gradient projection* method.



# Accelerated Proximal Gradient (APG) Method

Proximal gradient method can be combined with a momentum technique, Nesterov (1983), inspired by so-called *heavy ball* method by Polyak (1964).

Improved worst-case bound

$$f(x^k) - p^* = O(1/k^2)$$

One such method is *accelerated proximal gradient* method:

$$x^{k+1} = \mathbf{prox}_{\beta^{-1}h}(y^k - \beta^{-1}\nabla g(y^k))$$

$$\gamma_{k+1} = \frac{1 + \sqrt{1 + 4\gamma_k^2}}{2}$$

$$y^{k+1} = x^k + \frac{\gamma_k - 1}{\gamma_{k+1}}(x^k - x^{k-1})$$

with initial values  $\gamma_0 = 1$ ,  $y^{-1} = 0$ , and  $y^0 = x^0$ .

The APG method shown here is not a descent method.

## Second-Order Method

Consider minimize  $f(x)$  with additional assumption that  $f$  is twice differentiable.

Second-order Taylor expansion:

$$f(x + \Delta x) \approx f(x) + \nabla f(x)^T \Delta x + \frac{1}{2} \Delta x^T \nabla^2 f(x) \Delta x. \quad (155)$$

Derivative of right hand side with respect to  $\Delta x$  equal to zero:

$$\nabla^2 f(x) \Delta x = -\nabla f(x). \quad (156)$$

Compare with the first-order Taylor expansion of gradient:

$$\nabla f(x + \Delta x) \approx \nabla f(x) + \nabla^2 f(x) \Delta x$$

# Newton's Method

In the convex case Newton's method uses as search direction  $\Delta x = -\nabla^2 f(x)^{-1} \nabla f(x)$  which leads to the iteration

$$x^{k+1} = x^k - t_k \nabla^2 f(x^k)^{-1} \nabla f(x^k), \quad k = 0, 1, 2, \dots \quad (157)$$

where  $x^0$  is a initial guess, and  $t_k > 0$  is the step size at iteration  $k$ .

If  $x^k$  is sufficiently close to stationary point, then full Newton step can be shown to yield descent. However, this is not always the case if  $x^k$  is far away from stationary point.

## Backtracking Line Search

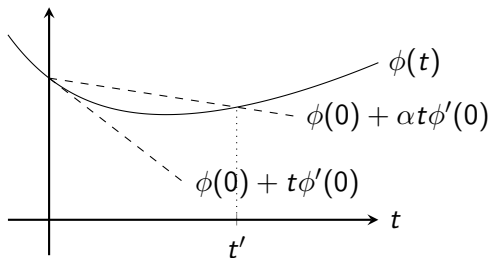
Given descent direction  $\Delta x$  at  $x^k$  consider  $\phi : \mathbf{R} \rightarrow \mathbf{R}$  defined as

$$\phi(t) = f(x^k + t\Delta x).$$

Since  $\Delta x$  descent direction, we have  $\phi'(0) = \nabla f(\Delta x)^T \Delta x < 0$ .

*Armijo* descent condition:

$$\phi(t) \leq \phi(0) + \alpha t \phi'(0). \quad (158)$$



# Backtracking Line Search Algorithm

**Require:** Initial step size  $t > 0$  and parameters  $\alpha \in (0, 1/2]$  and  $\beta \in (0, 1)$

**Ensure:**  $t > 0$  such that  $\phi(t) \leq \phi(0) + \alpha t \phi'(0)$

**while**  $\phi(t) > \phi(0) + \alpha t \phi'(0)$  **do**

$t \leftarrow \beta t$

**end while**

## Upper Bound on $\alpha$

Upper bound  $\alpha \leq 1/2$  can be motivated by considering case where

$$\phi(t) = \phi(0) + t\phi'(0) + \frac{t^2}{2}\phi''(0).$$

Exact minimizer of  $\phi(t)$  is then  $t^* = -\phi'(0)/\phi''(0)$ . The Armijo condition (158) imposes an upper bound on  $t$  which may be expressed in terms of  $t^*$  as  $t \leq 2(1 - \alpha)t^*$ .

It follows that  $t = t^*$  does not satisfy the descent condition if  $\alpha > 1/2$ .

# Trust-Region Method

In nonconvex case a trust-region is added:

$$\begin{array}{ll}\text{minimize} & \nabla f(x^k)^T \Delta x + \frac{1}{2} \Delta x^T \nabla^2 f(x^k) \Delta x \\ \text{subject to} & \|\Delta x\|_2^2 \leq \delta_k^2\end{array}$$

where  $\delta_k$  is the so-called trust-region radius.

Special case of a quadratically constrained quadratic problem (QCQP), and it can be solved numerically to global optimality by means of an eigenvalue decomposition of  $\nabla^2 f(x^k)$ .

# Equality Constrained Problems

Consider convex optimization problem

$$\begin{aligned} & \text{minimize} && f(x) \\ & \text{subject to} && Ax = b \end{aligned} \tag{159}$$

where  $x \in \mathbf{R}^n$ ,  $A \in \mathbf{R}^{m \times n}$  and  $b \in \mathbf{R}^m$  ( $A$  has full rank and  $m < n$ ).

Suppose  $F \in \mathbf{R}^{n \times p}$  basis for  $\mathcal{N}A$  and  $\bar{x}$  is any vector:  $A\bar{x} = b$ , then

$$\{x \in \mathbf{R}^n \mid Ax = b\} = \{Fz + \bar{x} \mid z \in \mathbf{R}^p\},$$

and hence (159) is equivalent to

$$\text{minimize}_z \quad f(Fz + \bar{x})$$

with variable  $z \in \mathbf{R}^p$ .

Newton equation:

$$F^T \nabla^2 f(Fz + \bar{x}) F \Delta z = -F^T \nabla f(Fz + \bar{x}).$$

Given a minimizer  $z^*$  of  $f(Fz + \bar{x})$ , a solution to the equality constrained problem may be obtained as  $x^* = Fz^* + \bar{x}$ .



# KKT Optimality Conditions

Lagrangian  $L : \mathbf{R}^n \times \mathbf{R}^m \rightarrow \mathbf{R}$  given by

$$L(x, \nu) = f(x) + \nu^T (Ax - b),$$

KKT conditions:

$$\begin{aligned}\nabla f(x) + A^T \nu &= 0 \\ Ax - b &= 0,\end{aligned}$$

or equivalently,  $r(z) = 0$  where  $z = (x, \nu)$  and

$$r(z) = \begin{bmatrix} \nabla f(x) + A^T \nu \\ Ax - b \end{bmatrix}.$$

## Infeasible start Newton method

Newton direction  $\Delta z = (\Delta x, \Delta \nu)$  for  $r(z) = 0$  solution to linearized equation:

$$r(z) + J_r(z)\Delta z = 0$$

where  $J_r$  is Jacobian of  $r$ . Equivalently

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta \nu \end{bmatrix} = - \begin{bmatrix} \nabla f(x) + A^T \nu \\ Ax - b \end{bmatrix}. \quad (160)$$

Step size using backtracking line search where

$$\phi(t) = \|r(z + t\Delta z)\|_2$$

is used with the Armijo condition  $\phi(t) \leq \phi(0) - \alpha t \phi'(0)$ . If  $r(z) \neq 0$  we have  $\phi'(0) = -\|r(z)\|_2$  and hence Armijo descent condition is

$$\|r(z + t\Delta z)\|_2 \leq (1 - t\alpha)\|r(z)\|_2. \quad (161)$$

## Variable Metric Methods

Consider more general convex quadratic approximation

$\hat{f}(\cdot | \cdot) : \mathbf{R}^n \times \mathbf{R}^n \rightarrow \mathbf{R}$  given by

$$\hat{f}(y | x^k) = f(x^k) + \nabla f(x^k)^T (y - x^k) + \frac{1}{2} (y - x^k)^T B_k (y - x^k) \quad (162)$$

where  $B_k \succ 0$ . We have

$$\hat{f}(x^k | x^k) = f(x^k), \quad \nabla \hat{f}(x^k | x^k) = \nabla f(x^k).$$

Define search direction as

$$\Delta x^k = \underset{\Delta x}{\operatorname{argmin}} \hat{f}(x^k + \Delta x | x^k) = -B_k^{-1} \nabla f(x^k),$$

and iterations as

$$x^{k+1} = x^k - t_k B_k^{-1} \nabla f(x^k). \quad (163)$$

Letting  $B_k = I$  yields the gradient descent method (144), and  $B_k = \nabla^2 f(x^k)$  corresponds to Newton's method.

## Secant Equation

The mechanism for updating  $B_k$  or  $H_k = B_k^{-1}$  is the *secant* equation

$$B_{k+1}(x^{k+1} - x^k) = \nabla f(x^{k+1}) - \nabla f(x^k). \quad (164)$$

Definition of  $\hat{f}$  implies  $\nabla \hat{f}(x^{k+1} | x^{k+1}) = \nabla f(x^{k+1})$ , and the secant equation is simply the additional condition

$$\nabla \hat{f}(x^k | x^{k+1}) = \nabla f(x^k)$$

Define  $y^k = \nabla f(x^{k+1}) - \nabla f(x^k)$  and  $s^k = x^{k+1} - x^k$  so that secant equation read

$$B_{k+1}s^k = y^k$$

Drop iteration index  $k$ .

## Quasi-Newton Update

The secant equation is an underdetermined for  $n > 1$ .

Quasi-Newton update formulae mostly differ in how they construct a matrix  $B_{k+1}$  that satisfies the secant equation.

The BFGS update, named after Broyden, Fletcher, Goldfarb, and Shanno, is

$$B_{k+1} = B_k - \frac{B_k s s^T B_k}{s^T B_k s} + \frac{y y^T}{y^T s}. \quad (165)$$

The update  $B_{k+1}$  is positive definite, and it can be shown to be the solution of convex optimization problem

$$\begin{array}{ll} \text{minimize} & \text{tr}(B B_k^{-1}) - \ln \det(B) \\ \text{subject to} & B s = y \end{array} \quad (166)$$

where  $B \in \mathbf{S}_+^n$ .

# Update Formula for Inverse

The BFGS update may also be expressed as

$$H_{k+1} = H_k + \frac{(y^T s + y^T H_k y) s s^T}{(y^T s)^2} - \frac{H_k y s^T + s y^T H_k}{y^T s} \quad (167)$$

where  $H_{k+1} = B_{k+1}^{-1}$ .

Hence

$$\Delta x = -H_k \nabla f(x^k)$$

# Davidson, Fletcher, and Powell (DFP) update

DFP update:

$$B_{k+1} = B_k + \frac{ry^T + yr^T}{y^T s} - \frac{(r^T s)yy^T}{(y^T s)^2} \quad (168)$$

where  $r = y - B_k s$ , or equivalently, as inverse Hessian approximation  $H_{k+1} = B_k^{-1}$ ,

$$H_{k+1} = H_k - \frac{H_k y y^T H_k}{y^T H_k y} + \frac{s s^T}{y^T s}. \quad (169)$$

## Symmetric Rank-1 (SR1) Update

SR1 update:

$$B_{k+1} = B_k + \sigma vv^T$$

where  $\sigma \in \{-1, 1\}$ . The secant equation  $B_{k+1}s = y$  implies

$$r = \sigma(v^T s)v \quad (170)$$

where  $r = y - B_k s$ . Forming the inner product with  $s$  on both sides of this equation yields  $\sigma(v^T s)^2 = r^T s$  which, combined with (170), implies

$$vv^T = \frac{rr^T}{\sigma^2(v^T s)^2} = \frac{rr^T}{\sigma r^T s},$$

and hence

$$B_{k+1} = B_k + \frac{rr^T}{r^T s}. \quad (171)$$

Update maintains symmetry, but unlike BFGS and DFP updates, it does not guarantee that  $B_{k+1}$  is positive definite.



## Barzilai–Borwein (BB) Step Size Rule

If we require  $B_{k+1} = \gamma_{k+1}I$  where  $\gamma_{k+1} \in \mathbf{R}_{++}$ , the secant condition is generally not satisfiable, but one can choose  $\gamma_{k+1}$  such that

$$\gamma_{k+1} = \operatorname{argmin}_{\gamma > 0} \|\gamma s - y\|_2^2 = \frac{y^T s}{\|s\|_2^2}. \quad (172)$$

Alternatively, we can define an inverse approximation  $H_{k+1} = \gamma_{k+1}I$  and get

$$\gamma_{k+1} = \operatorname{argmin}_{\gamma > 0} \|s - \gamma y\|_2^2 = \frac{y^T s}{\|y\|_2^2}. \quad (173)$$

Combining the gradient descent method with one of the two BB step size rules does not yield a descent method. The method converges if  $f$  is quadratic, but it is not guaranteed to converge in general. However, convergence can be established if the BB steps are combined with a line search.

## Proximal Quasi-Newton

Variable metric approach can be extended to  $f = g + h$  where  $h$  is convex and  $g$  is twice differentiable and convex. Let

$$\hat{f}(y | x^k) = h(y) + g(x^k) + \nabla g(x^k)^T (y - x^k) + \frac{1}{2} (y - x^k)^T B_k (y - x^k) \quad (174)$$

and let

$$\begin{aligned} x^{k+1} &= \operatorname{argmin}_y \left\{ \hat{f}(y | x^k) \right\} \\ &= \operatorname{argmin}_y \left\{ h(y) + \frac{1}{2} \|y - (x^k - B_k^{-1} \nabla g(x^k))\|_{B_k}^2 \right\} \\ &= \mathbf{prox}_{h^{B_k}}^{B_k}(x^k - B_k^{-1} \nabla g(x^k)) \end{aligned} \quad (175)$$

where

$$\mathbf{prox}_h^B(x) = \operatorname{argmin}_y \left\{ h(y) + \frac{1}{2} \|y - x\|_B^2 \right\}. \quad (176)$$

The iteration (175) is known as *proximal quasi-Newton* or *proximal Newton* if  $B_k = \nabla^2 f(x^k)$ .

# Stochastic Optimization Methods

Consider *stochastic optimization* problem

$$\text{minimize } \mathbb{E}_{\xi}[F(x, \xi)] \quad (177)$$

with variable  $x \in \mathbf{R}^n$  and where  $\xi$  is a random variable.<sup>5</sup>

In statistical learning,  $\xi$  typically represents the problem data, *i.e.*, each realization of  $\xi$  corresponds to a random observation.

The problem (177) is reduced to a deterministic problem if we let  $f(x) = \mathbb{E}_{\xi}[F(x, \xi)]$ .

However, the probability distribution associated with the random variable  $\xi$  is required to compute the expectation, and this distribution is typically not available in practice.

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<sup>5</sup>We refrain from a stringent mathematical definition in order not to clutter the presentation.

# Sample Average Approximation (SAA)

Replace the expectation by

$$\mathbb{E}_{\xi}[F(x, \xi)] \approx \frac{1}{m} \sum_{i=1}^m F(x, \xi_i)$$

where  $\xi_1, \dots, \xi_m$  are  $m$  independent samples of the random variable  $\xi$ .

Resulting problem is deterministic:

$$\text{minimize} \quad \frac{1}{m} \sum_{i=1}^m f_i(x) \quad (178)$$

where  $f_i(x) = F(x, \xi_i)$ .

An example is *empirical risk minimization* where

$$f_i(x) = l(a_i^T x, b_i)$$

where  $l: \mathbf{R}^n \rightarrow \mathbf{R}$  is a loss function and  $(a_i, b_i) \in \mathbf{R}^n \times \mathbf{R}$  is one of  $m$  observations.

# Stochastic Approximation (SA) Method

Find a root of  $\nabla f(x) = 0$  where  $f(x) = \mathbb{E}_\xi[F(x, \xi)]$ .

Algorithm:

$$x^{k+1} = x^k - t_k g^k \quad (179)$$

where  $t_k > 0$  step size and  $g^k$  is an unbiased estimator of  $\nabla f(x^k)$  with bounded variance, *i.e.*,

$$\mathbb{E}[g^k] = \nabla f(x^k), \quad \mathbb{E}[\|g^k\|^2] \leq c$$

for all  $k$ .

If  $\xi$  is independent of  $x$ , then  $\mathbb{E}_\xi[\nabla F(x, \xi)] = \nabla f(x)$ , and hence  $g^k = \nabla F(x^k, \xi_k)$  is an unbiased estimator of  $\nabla f(x^k)$  if  $\xi_k$  is a random realization of  $\xi$ .

Convergence to a stationary point requires a step size sequence:

$$t_k \geq 0, \quad \lim_{k \rightarrow \infty} t_k = 0, \quad \sum_{k=0}^{\infty} t_k = \infty.$$

## Some Comments

An example step size sequence:

$$t_k = \frac{t}{(k+1+c)^\alpha}, \quad k = 0, 1, 2, \dots$$

where  $t > 0$ ,  $c \geq 0$ , and  $\alpha \in (1/2, 1]$  are parameters.

Choosing  $t_k$  large may result in diverging behavior in early iterations, whereas choosing  $t_k$  small often results in poor performance in later iterations. The parameter  $c$  can trade off this.

The iteration (179) is often referred to as a *stochastic gradient* (SG) method or a *stochastic gradient descent* (SGD) method.

The stochastic gradient  $g^k$  is not necessarily a descent direction.

We note that in some areas, the step size  $t_k$  is referred to as the *learning rate*.

# Incremental Methods

The SG method is closely related to the class of *incremental methods*, and the two terms are often used synonymously.

Incremental methods solves (178) using

$$x^{k+1} = x^k - t_k \nabla f_{i_k}(x^k)$$

where  $i_k \in \mathbf{N}_m$  is chosen according to some index selection rule.

Two of most common rules are the *cyclic* rule  $i_k = (k \bmod m) + 1$  and the *random* rule where each  $i_k$  is chosen uniformly at random from  $\mathbf{N}_m$ .

With random index selection rule, incremental gradient method may be viewed as SG method applied to (177) if random variable  $\xi$  is discrete with  $m$  equiprobable outcomes.

## Batch Gradient Method

Consider again

$$\text{minimize } \frac{1}{m} \sum_{i=1}^m f_i(x) \quad (180)$$

which has gradient

$$\nabla f(x^k) = \frac{1}{m} \sum_{i=1}^m \nabla f_i(x^k).$$

If full gradient is used the gradient method is often referred to as *batch* gradient descent.

The *incremental* gradient method only uses

$$\nabla f_{i_k}(x^k)$$

in each iteration

Can be viewed as noisy approximation of full gradient and is not necessarily a descent direction.



# Mini-Batch Gradient Method

Compromise:

$$x^{k+1} = x^k - \frac{1}{p} \sum_{i \in \mathcal{S}_k} \nabla f_i(x^k)$$

where  $\mathcal{S}_k \subset \mathbf{N}_m$  and  $|\mathcal{S}_k| = p$ .

# Adaptive Step Size

Diminishing step size sequence required for convergence.

Adaptive step size rules needed for practical convergence.

Prototype method:

$$x^{k+1} = x^k - B_k^{-1} g^k$$

where  $B_k \in \mathbf{S}_{++}^n$  and  $g^k$  is stochastic gradient.

Interpretations: incremental/stochastic variant of variable metric method, or adaptively preconditioned stochastic gradient method.

# Adaptive Gradient Method (AdaGrad)

Consider again

$$\text{minimize } \mathbb{E}_{\xi}[F(x, \xi)] \quad (181)$$

with variable  $x \in \mathbf{R}^n$  and where  $\xi$  is a random variable.

Assume  $F(x, \xi) = G(x, \xi) + h(x)$  with  $G(\cdot, \xi)$  and  $h$  closed and convex.

Let  $g^k \in \partial G(x^k, \xi_k)$  where  $\xi_k$  a realization of  $\xi$  at iteration  $k$  and

$$\hat{G}_k = \sum_{i=0}^k g^i (g^i)^T,$$

AdaGrad uses either  $B_k = \gamma(\text{diag}(\hat{G}_k) + \varepsilon I)^{1/2}$  or  $B_k = \gamma(\hat{G}_k + \varepsilon I)^{1/2}$  where  $\gamma > 0$  and  $\varepsilon > 0$  are parameters.

# Diagonal AdaGrad

Diagonal variant of AdaGrad:

$$v^k = v^{k-1} + g^k \circ g^k$$

$$B_k = \gamma \mathbf{diag}(v^k + \varepsilon I)^{1/2}$$

$$x^{k+1} = \mathbf{prox}_h^{B_k}(x^k - B_k^{-1} g^k)$$

where  $v^{-1} = 0$  and  $x^0$  are initial values.

# Implementational Details

The matrix  $B_k$  can be expressed as

$$B_k = \sqrt{k+1} \left( \frac{1}{k+1} \mathbf{diag} \hat{G}_k + \frac{\varepsilon}{k+1} I \right)^{1/2}$$

which may be implemented recursively as

$$\begin{aligned} \tilde{v}^k &= \frac{k}{k+1} \tilde{v}^{k-1} + \frac{1}{k+1} (g^k \circ g^k) \\ B_k &= \gamma \sqrt{k+1} \mathbf{diag}(\tilde{v}^k + \varepsilon/(k+1) \mathbf{1})^{1/2}. \end{aligned}$$

# Summary of AdaGrad

Then

$$\begin{aligned}\tilde{v}^k &= \frac{k}{k+1} \tilde{v}^{k-1} + \frac{1}{k+1} (g^k \circ g^k) \\ B_k &= \gamma \sqrt{k+1} \mathbf{diag}(\tilde{v}^k + \varepsilon/(k+1) \mathbf{1})^{1/2} \\ \tilde{g}^k &= \mathbf{diag}(\tilde{v}^k + \varepsilon/(k+1) \mathbf{1})^{-1/2} g^k \\ x^{k+1} &= \mathbf{prox}_h^{B_k}(x^k - t_k \tilde{g}^k)\end{aligned}$$

with  $t_k = \frac{1}{\gamma \sqrt{k+1}}$ , which shows that AdaGrad implicitly employs a diminishing step size sequence.

In the convex setting, AdaGrad can be shown to satisfy the worst-case bound

$$\mathbb{E}[f(x^k) - p^*] \leq O(1/\sqrt{k}).$$

# RMSprop

Consider again

$$\text{minimize } \mathbb{E}_{\xi}[F(x, \xi)] \quad (182)$$

with variable  $x \in \mathbf{R}^n$  and where  $\xi$  is a random variable. Assume  $F(x, \xi)$  differentiable with gradient  $g^k = \nabla F(x^k, \xi_k)$ .

Stochastic gradient iteration using moving average

$$\begin{aligned} v^k &= \beta v^{k-1} + (1 - \beta)(g^k \circ g^k) \\ B_k &= \gamma \text{diag}(v^k + \varepsilon \mathbf{1})^{1/2} \\ x^{k+1} &= x^k - B_k^{-1} g^k \end{aligned}$$

where  $\beta \in (0, 1)$ .

RMSprop does not implicitly result in a diminishing step size sequence, and in fact, the method need not converge.

# AdaDelta

Same optimization problem as before.

AdaDelta (similar to RMSprop but more averaging):

$$v^k = \beta v^{k-1} + (1 - \beta)(g^k \circ g^k)$$

$$B_k = \mathbf{diag}(v^k + \varepsilon \mathbf{1})^{1/2} \mathbf{diag}(p^{k-1} + \varepsilon \mathbf{1})^{-1/2}$$

$$p^k = \beta p^{k-1} + (1 - \beta) \left( B_k^{-1} g^k \circ B_k^{-1} g^k \right)$$

$$x^{k+1} = x^k - B_k^{-1} g^k.$$

We note that AdaDelta need not converge.



# Adaptive Moment Estimation (Adam)

Combines the adaptive scaling approach of RMSprop with *gradient aggregation* or *momentum*.

$$\mu^k = \beta_1 \mu^{k-1} + (1 - \beta_1) g^k$$

$$v^k = \beta_2 v^{k-1} + (1 - \beta_2)(g^k \circ g^k)$$

$$H_k = \gamma^{-1} \frac{\sqrt{1 - \beta_2^k}}{1 - \beta_1^k} \mathbf{diag}(\mu^k) \left( \mathbf{diag}(v^k)^{1/2} + \varepsilon I \right)^{-1}$$

$$x^{k+1} = x^k - H_k \mu^k.$$

The method need not converge.

Variant called *AdaMax* obtained by replacing  $v^k$  and  $H_k$  by

$$u^k = \max(\beta_2 u^{k-1}, |g^k|), \quad H_k = \gamma^{-1} \frac{1}{1 - \beta_1^k} \mathbf{diag}(u^k)^{-1}$$

*Nadam* combines Adam with Nesterov-like acceleration.

Addresses Adam's convergence issue by avoiding the possibility of increasing step sizes:

$$\mu^k = \beta_{1,k}\mu^{k-1} + (1 - \beta_{1,k})g^k$$

$$v^k = \beta_2 v^{k-1} + (1 - \beta_2)(g^k \circ g^k)$$

$$\hat{v}^k = \max(\hat{v}^{k-1}, v^k)$$

$$B_k = \gamma\sqrt{k+1} \mathbf{diag}(\hat{v}^k)^{1/2}$$

$$x^{k+1} = \mathbf{prox}_h^{B_k}(x^k - B_k^{-1}\mu^k)$$

where  $\hat{v}^k = \max(\hat{v}^{k-1}, v^k)$  denotes the elementwise maximum of  $\hat{v}^{k-1}$  and  $v^k$ .

Adaptive method based on technique called *weight normalization*.

Method can be used with both gradient method and stochastic gradient method.

$$x^{k+1} = x^k - t_k g^k$$
$$t_{k+1} = \frac{t_k}{1 + \|t_k g^k\|_2^2}$$

where  $t_0 > 0$ , and  $g^k$  is either stochastic gradient or gradient of  $f$ .

In batch setting (i.e.,  $g^k = \nabla f(x^k)$ ), the method converges globally to stationary point if  $f$  is  $\beta$ -smooth, and in stochastic setting (i.e.,  $g^k = \nabla F(x^k, \xi_k)$ ), convergence for convex function.

# Augmented Lagrangian Methods

Consider

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & h(x) = 0 \end{array} \quad (183)$$

where  $f_0 : \mathbf{R}^n \rightarrow \mathbf{R}$ , and where  $h : \mathbf{R}^n \rightarrow \mathbf{R}^p$ .

Equivalent problem

$$\begin{array}{ll} \text{minimize} & f_0(x) + \frac{\rho}{2} \|h(x)\|_2^2 \\ \text{subject to} & h(x) = 0 \end{array} \quad (184)$$

where  $\rho \in \mathbf{R}_{++}$  called *penalty parameter*.

# Augmented Lagrangian

Lagrangian for equivalent problem  $L_\rho : \mathbf{R}^n \times \mathbf{R}^p \rightarrow \mathbf{R}$ , called *augmented Lagrangian*

$$L_\rho(x, \mu) = f_0(x) + \mu^T h(x) + \frac{\rho}{2} \|h(x)\|_2^2$$

where  $\rho \in \mathbf{R}_{++}$ .

Dual function  $g_\rho : \mathbf{R}^p \rightarrow \mathbf{R}$

$$g_\rho(\mu) = \inf_x L_\rho(x, \mu)$$

We will only discuss first order methods.

# Dual Ascent

Use a gradient method to maximize  $g_\rho$ .

The solution to the original problem can be found from the optimal dual variable  $\mu^*$  by solving

$$\inf_x L_\rho(x, \mu^*)$$

Key to dual ascent is to be able to compute a gradient or at least a sub-gradient of  $g_\rho$ .

## Sub-Gradient

Let  $(x^k, \mu^k) \in \mathbf{R}^n \times \mathbf{R}^p$  be current iterate and define next iterate  $x^{k+1} \in \mathbf{R}^n$  as the solution of

$$\inf_x L_\rho(x, \mu^k)$$

Then

$$\begin{aligned} g_\rho(\mu) &= \inf_x L_\rho(x, \mu) \\ &= \inf_x \left( f_0(x) + \mu^T h(x) + \frac{\rho}{2} \|h(x)\|_2^2 \right) \\ &\leq f_0(x^{k+1}) + \mu^T h(x^{k+1}) + \frac{\rho}{2} \|h(x^{k+1})\|_2^2 \\ &= f_0(x^{k+1}) + (\mu^k)^T h(x^{k+1}) + (\mu - \mu^k)^T h(x^{k+1}) + \frac{\rho}{2} \|h(x^{k+1})\|_2^2 \\ &= g_\rho(\mu^k) + (\mu - \mu^k)^T h(x^{k+1}) \end{aligned}$$

and hence  $h(x^{k+1})$  is a sub-gradient of  $g_\rho$  at  $\mu^k$ .

# Differentiable Case

If differentiability

$$0 = \nabla L_\rho(x^{k+1}, \mu^k) = \nabla f_0(x^{k+1}) + \nabla h(x^{k+1})^T (\mu^k + \rho h(x^{k+1}))$$

and natural update for  $\mu$  is

$$\mu^{k+1} = \mu^k + \rho h(x^{k+1})$$

since this is dual ascent step which also makes  $\mu^{k+1}$  dual feasible, i.e.

$$\nabla f_0(x^{k+1}) + \nabla h(x^{k+1})^T \mu^{k+1} = 0$$



# Method of Multipliers

Given current iterate  $(x^k, \mu^k)$  update as:

1.  $x^{k+1} = \operatorname{argmin}_x L_\rho(x, \mu^k)$
2.  $\mu^{k+1} = \mu^k + \rho h(x^{k+1})$

It possible to also update the parameter  $\rho$ :

$$\rho^{k+1} = \begin{cases} \beta \rho^k, & \|h(x^{k+1})\|_2 > \gamma \|h(x^k)\|_2 \\ \rho_k, & \|h(x^{k+1})\|_2 \leq \gamma \|h(x^k)\|_2 \end{cases}$$

where  $\beta > 0$ .

# Linear Equality Constraints

Consider

$$\begin{array}{ll} \text{minimize} & F(x) + G(y) \\ \text{subject to} & Ax + By = c \end{array} \quad (185)$$

where  $F : \mathbf{R}^n \rightarrow \mathbf{R}$ ,  $G : \mathbf{R}^m \rightarrow \mathbf{R}$ ,  $A \in \mathbf{R}^{p \times n}$ ,  $B \in \mathbf{R}^{p \times m}$ , and  $c \in \mathbf{R}^p$ .

We assume  $F$  and  $G$  convex but not necessarily differentiable.

Augmented Lagrangian  $L_\rho : \mathbf{R}^n \times \mathbf{R}^m \times \mathbf{R}^p \rightarrow \mathbf{R}$ :

$$L_\rho(x, y, \mu) = F(x) + G(y) + \mu^T (Ax + By - c) + \frac{\rho}{2} \|Ax + By - c\|_2^2$$

# Alternating Direction Methods of Multipliers (ADMM)

Block-coordinate minimization:

1.  $x^{k+1} = \operatorname{argmin}_x L_\rho(x, y^k, \mu^k)$
2.  $y^{k+1} = \operatorname{argmin}_y L_\rho(x^{k+1}, y, \mu^k)$
3.  $\mu^{k+1} = \mu^k + \rho (Ax^{k+1} + By^{k+1} - c)$

Also  $\rho$  can be updated.

Often the minimizations in 1. and 2. are easier to carry out than the joint minimization in the method of multipliers.

# Equivalent Formulation

Let  $u^k = \mu^k / \rho$ . Then equivalent scaled version of ADMM is

1.  $x^{k+1} = \operatorname{argmin}_x (F(x) + \frac{\rho}{2} \|Ax + By^k - c + u^k\|_2^2)$
2.  $y^{k+1} = \operatorname{argmin}_y (G(y) + \frac{\rho}{2} \|Ax^{k+1} + By - c + u^k\|_2^2)$
3.  $u^{k+1} = u^k + Ax^{k+1} + By^{k+1} - c$

When  $A = I$  we have

$$x^{k+1} = \mathbf{prox}_{\rho^{-1}F} (c - u^k - By^k)$$

A similar result holds for 2. when  $B = I$ .

# Common Form of Optimization Problem

Consider

$$\text{minimize}_x F(x) + G(x)$$

where  $G(x) = \lambda \|x\|$  for some norm  $\|\cdot\| : \mathbf{R}^m \rightarrow \mathbf{R}_+$  and where  $\lambda \in \mathbf{R}_{++}$ .

Equivalently

$$\begin{array}{ll} \text{minimize}_{x,y} & F(x) + G(y) \\ \text{subject to} & x - y = 0 \end{array} \quad (186)$$

where  $y \in \mathbf{R}^n$ .

ADMM updates

1.  $x^{k+1} = \mathbf{prox}_{\rho^{-1}F}(y^k - u^k)$
2.  $y^{k+1} = \mathbf{prox}_{\rho^{-1}G}(x^{k+1} + u^k)$
3.  $u^{k+1} = u^k + x^{k+1} - y^{k+1}$

# Proximal Operators

For 1-norm  $f(x) = \sum_{i=1}^m |x_i|$  has

$$\mathbf{prox}_f(x)_i = \begin{cases} x_i - 1, & x_i \geq 1 \\ 0, & |x_i| \leq 1 \\ x_i + 1, & x_i \leq -1 \end{cases}$$

For nuclear norm of a matrix  $X$ , i.e.  $f(X) = \|X\|_*$

$$\mathbf{prox}_f(X) = \sum_{i=1}^m (\sigma_i - 1)_+ u_i v_i^T$$

where  $X = \sum_{i=1}^m \sigma_i u_i v_i^T$  is an SVD of  $X$  and where  $(\cdot)_+ : \mathbf{R} \rightarrow \mathbf{R}$  is the function defined by

$$(z)_+ = \begin{cases} z, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

# Scaling Rule

If  $f : \mathbf{R}^m \rightarrow \mathbf{R}$  and  $g : \mathbf{R}^n \rightarrow \mathbf{R}$  are functions related as  $f(x) = \lambda g(x/\lambda)$ , for  $\lambda \in \mathbf{R}_{++}$ , then

$$\mathbf{prox}_f(x) = \lambda \mathbf{prox}_{\lambda^{-1}g}(x/\lambda)$$

# Sum of Functions

Assume  $F_i : \mathbf{R}^n \rightarrow \mathbf{R}$  and

$$F(x) = \sum_{i=1}^N F_i(x)$$

Then equivalent optimization problem is

$$\begin{array}{ll} \text{minimize}_{x_1, \dots, x_N, y} & \sum_{i=1}^N F_i(x_i) + G(y) \\ \text{subject to} & x_i - y = 0, \quad i \in \mathbf{N}_N \end{array} \quad (187)$$

where  $x_i \in \mathbf{R}^n$

First update is

$$x_i^{k+1} = \text{prox}_{\rho^{-1}F}(y^k - u_i^k), \quad i \in \mathbf{N}_N$$

with  $u = (u_1, \dots, u_N)$ .



# Completing Squares

Second update is

$$\text{minimize}_y G(y) + \frac{\rho}{2} \sum_{i=1}^N \left\| x_i^{k+1} - y + u_i^k \right\|_2^2$$

Expand the squared norm expressions, complete the squares with respect to  $y$ , and ignore terms independent of  $y$ :

$$\text{minimize}_y G(y) + \frac{N\rho}{2} \left\| y - \left( \bar{x}^{k+1} + \bar{u}^k \right) \right\|_2^2$$

where

$$\bar{x}^{k+1} = \frac{1}{N} \sum_{i=1}^N x_i^{k+1}; \quad \bar{u}^k = \frac{1}{N} \sum_{i=1}^N u_i^k$$

# Distributed ADMM

ADMM algorithm distributes:

1.  $x_i^{k+1} = \mathbf{prox}_{\rho^{-1}F}(y^k - u_i^k), \quad i \in \mathbf{N}_N$
2.  $\bar{x}^{k+1} = \frac{1}{N} \sum_{i=1}^N x_i^{k+1}$
3.  $\bar{u}^k = \frac{1}{N} \sum_{i=1}^N u_i^k$
4.  $y^{k+1} = \mathbf{prox}_{(N\rho)^{-1}G}(\bar{x}^{k+1} + \bar{u}^k)$
5.  $u_i^{k+1} = u_i^k + x_i^{k+1} - y^{k+1}, \quad i \in \mathbf{N}_N$

Notice that 1. and 5. can be carried out in parallel.

# General Constrained Optimization Problem

Consider

$$\begin{array}{ll} \text{minimize} & F(x) \\ \text{subject to} & x \in C \end{array} \quad (188)$$

where  $F : \mathbf{R}^n \rightarrow \mathbf{R}$  convex function for which proximal operator can be evaluated cheaply, and where  $C$  is convex set.

Equivalent problem

$$\begin{array}{ll} \text{minimize} & F(x) + \delta_C(y) \\ \text{subject to} & x - y = 0. \end{array} \quad (189)$$

The prox operator of the indicator function of a convex set obtained from

$$\text{minimize}_{y \in C} \|y - x\|_2^2$$

which is the projection of  $x$  onto  $C$ .

# Barrier Problem

Consider

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i \in \mathbf{N}_m \\ & Ax = b \end{array} \quad (190)$$

where  $f_0 : \mathbf{R}^n \rightarrow \mathbf{R}$ ,  $f_i : \mathbf{R}^n \rightarrow \mathbf{R}$ ,  $i \in \mathbf{N}_m$  are convex functions, and where  $A \in \mathbf{R}^{p \times n}$  and  $b \in \mathbf{R}^p$ .

Replace inequality constraints with *barrier function*  $\phi : \mathbf{R}^n \rightarrow \mathbf{R}$  where

$$\phi(x) = - \sum_{i=1}^m \ln(-f_i(x))$$

Barrier problem:

$$\begin{array}{ll} \text{minimize} & f_0(x) + \frac{1}{t} \phi(x) \\ \text{subject to} & Ax = b \end{array} \quad (191)$$

where  $t > 0$ .

# Optimality Conditions

We have

$$\nabla \phi(x) = \sum_{i=1}^m \frac{1}{-f_i(x)} \nabla f_i(x)$$

$$\nabla^2 \phi(x) = \sum_{i=1}^m \frac{1}{f_i(x)^2} \nabla f_i(x) \nabla f_i(x)^T + \sum_{i=1}^m \frac{1}{-f_i(x)} \nabla^2 f_i(x)$$

Lagrangian  $L : \mathbf{R}^n \times \mathbf{R}^p$  of (191) defined by

$$L(x, \lambda) = f_0(x) + \frac{1}{t} \phi(x) + \mu^T (Ax - b)$$

Hence optimality conditions for (191) are

$$\nabla f_0(x) - \sum_{i=1}^m \frac{1}{tf_i(x)} \nabla f_i(x) + A^T \mu = 0$$

$$Ax = b$$

# Interior-Point (IP) Method

Conditions can be solved with Newton's method by linearization.

Equations for search directions  $(\Delta x, \Delta \mu) \in \mathbf{R}^x \times \mathbf{R}^p$ :

$$\left( \nabla^2 f_0(x) + \frac{1}{t} \nabla^2 \phi(x) \right) \Delta x + A^T \Delta \mu = -\nabla f_0(x) - \frac{1}{t} \nabla \phi(x) - A^T \mu$$
$$A \Delta x = -(Ax - b)$$

where  $(x, \mu)$  is the current iterate.

Use Newton's method to solve the barrier problem, increase the value of  $t$  and repeat.

## Termination Criterion

Define the Lagrangian  $\mathcal{L} : \mathbf{R}^n \times^m \times \mathbf{R}^p \rightarrow \mathbf{R}$  for (190) as

$$\mathcal{L}(x, \lambda, \mu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \mu^T (Ax - b)$$

If  $x^*$  solves barrier problem, then by optimality conditions above for (191) there exist  $\mu$ :

$$\nabla f_0(x^*) - \sum_{i=1}^m \frac{1}{tf_i(x^*)} \nabla f_i(x^*) + A^T \mu = 0$$

and hence  $x^*$  minimizes  $\mathcal{L}(x, \lambda^*, \mu)$  where  $\lambda_i^* = 1/(-tf_i(x^*)) \geq 0$ .

Let  $g : \mathbf{R}^m \times \mathbf{R}^p$  be Lagrange dual function, i.e.

$g(\lambda, \mu) = \min_x \mathcal{L}(x, \lambda, \mu)$ . By weak duality

$$p^* \geq g(\lambda^*, \mu) = \mathcal{L}(x^*, \lambda^*, \mu) = f_0(x^*) - m/t$$

Above  $p^*$  is the optimal value of (190). Hence  $x^*$  from the barrier problem is  $\epsilon$ -suboptimal with  $\epsilon = m/t$ .

# Strictly Feasible Initial Point

*Phase I* problem:

$$\begin{array}{ll}\text{minimize} & s \\ \text{subject to} & f_i(x) \leq s, \quad i \in \mathbf{N}_m \\ & Ax = b\end{array}$$

in the variables  $(x, s) \in \mathbf{R}^{n+1}$ .

If optimal solution of this problem has  $s^* < 0$ , then corresponding  $x^*$  is strictly feasible for (190).



## Other IP Methods

Efficient IP methods update the primal and dual variables jointly and are called *primal-dual* IP methods.

Specifically the corresponding barrier problem is not solved to optimality before  $t$  is updated.

## Graphical Normal Distribution

The dual problem related to graphical normal distribution:

$$\begin{array}{ll} \text{minimize} & \text{tr } CX - \ln \det X \\ \text{subject to} & \text{tr } A_i X = b_i, \quad i \in \mathbf{N}_p \end{array} \quad (192)$$

where  $X \in \mathbf{S}_+^n$ ,  $A_i \in \mathbf{S}^n$ ,  $b_i \in \mathbf{R}$ , and  $C \in \mathbf{S}^n$ .

Lagrangian  $L : \mathbf{S}_+^n \times \mathbf{R}^p$  given by

$$L(X, y) = \text{tr } CX - \ln \det X + \sum_{i=1}^p y_i (\text{tr } A_i X - b_i)$$

Optimality conditions

$$\begin{aligned} C - X^{-1} + \sum_{i=1}^p y_i A_i &= 0 \\ \text{tr } A_i X - b_i &= 0, \quad i \in \mathbf{N}_p \end{aligned}$$

# Equivalent Optimality Conditions

Let  $X = E(x) = \sum_{i=1}^m x_i E_i$  where  $E_i$  are basis matrices for a symmetric  $n \times n$  matrix,  $m = n(n+1)/2$  and where  $E : \mathbf{R}^m \rightarrow \mathbf{S}^n$  and  $x = (x_1, \dots, x_m)$ .

Equivalent optimality conditions

$$\begin{aligned} \text{tr } CE_k - \text{tr } E(x)^{-1} E_k + \sum_{i=1}^p y_i \text{tr } A_i E_k &= 0, & k \in \mathbf{N}_m \\ \text{tr } A_i X - b_i &= 0, & i \in \mathbf{N}_p \end{aligned}$$

# Newton's Method

Conditions can be solved with Newton's method by linearization.

Equations for search directions  $(\Delta x, \Delta y) \in \mathbf{R}^m \times \mathbf{R}^p$ :

$$\sum_{j=1}^m \text{tr } E_k X^{-1} E_j X^{-1} \Delta x_j + \sum_{i=1}^p \text{tr } A_i E_k \Delta y_i = -\text{tr } C E_k + \text{tr } X^{-1} E_k$$
$$- \sum_{i=1}^p y_i \text{tr } A_i E_k, \quad k \in \mathbf{N}_m$$
$$\sum_{j=1}^m \text{tr } A_i E_j \Delta x_j = -\text{tr } A_i X + b_i, \quad i \in \mathbf{N}_p$$

where  $(X, y)$  is the current iterate.

Cost of solving equations minor compared to forming the equations.

# Conic Optimization Problem

Consider

$$\begin{array}{ll} \text{minimize} & \mathbf{tr} CX \\ \text{subject to} & \mathbf{tr} A_i X = b_i, \quad i \in \mathbf{N}_p \\ & X \in \mathbf{S}_+^n \end{array} \quad (193)$$

Lagrangian  $\mathcal{L} : \mathbf{S}^n \times \mathbf{S}^n \times \mathbf{R}^p$  is

$$\mathcal{L}(X, Z, y) = \mathbf{tr} CX - \mathbf{tr} ZX + \sum_{i=1}^p y_i (\mathbf{tr} A_i X - b_i)$$

# Lagrange Dual Problem

We have

$$\mathcal{L}(X, Z, y) = \mathbf{tr} \left( C + \sum_{i=1}^p y_i A_i - Z \right) X - b^T y$$

where  $y = (y_1, \dots, y_p)$  and  $b = (b_1, \dots, b_p)$ .

Lagrange dual problem

$$\begin{array}{ll} \text{maximize} & -b^T y \\ \text{subject to} & C + \sum_{i=1}^p y_i A_i = Z \\ & Z \in \mathbf{S}_+^n \end{array} \quad (194)$$

## Barrier Function

Let  $X^*$  be optimal for (192). Then  $X^*$  primal feasible and there  $\exists y$ :

$$C - (X^*)^{-1} + \sum_{i=1}^p y_i A_i = 0$$

which is first row of optimality conditions for (192).

Moreover  $Z = (X^*)^{-1}$  is dual feasible since  $X^* \in \mathbf{S}_{++}^n$ . Hence by the optimality conditions for (192) the duality gap is

$$\begin{aligned} \text{tr } CX^* + b^T y &= \text{tr} \left( (X^*)^{-1} - \sum_{i=1}^p y_i A_i \right) X^* + b^T y \\ &= n - \sum_{i=1}^p y_i \text{tr } A_i X^* + b^T y = n \end{aligned}$$

If  $\ln \det X$  in (192) replaced with  $\frac{1}{t} \ln \det X$  duality gap is  $n/t$ , showing that  $\ln \det X$  barrier function.

# Logarithmic Barrier Function

Function  $\psi : K \rightarrow \mathbf{R}$ , called *generalized logarithm* for a proper cone  $K$  if

1.  $\text{dom } \psi = \text{int} K$
2.  $\nabla \psi(x) \in K^* \ \forall x \in K$
3.  $\exists \theta$  such that  $\psi(sx) = \psi(x) + \theta \ln s \ \forall x \in K$  and  $s > 0$ .

Here  $\theta$  is called the degree of  $\psi$ .

For any generalized logarithm it holds that for  $x \in K$  we have  $\langle x, \nabla \psi(x) \rangle = \theta$ . The duality gap will be  $\theta/t$  for solutions from the barrier problem.

Generalized logarithms for different cones

- ▶  $\psi(x) = \sum_{i=1}^n \ln(x_i)$ , where  $x = (x_1, \dots, x_n) \in \mathbf{R}_+^n$  with  $\theta = n$ .
- ▶  $\psi(X) = \ln \det X$ , where  $X \in \mathbf{S}_+^n$  with  $\theta = n$ .
- ▶  $\psi(x) = \ln(y_n^2 - y_1^2 - \dots - y_{n-1}^2)$ , where  $x \in \mathbf{Q}^n$  with  $\theta = 2$ .
- ▶  $\psi(x) = \ln(y \ln(z/y) - x) + \ln y + \ln z$ , where  $(x, y, z) \in K_{\text{exp}}$  with  $\theta = 3$ .